

A clearing fuel market for AeroMAPS

Step 1: formulation, integration, verification and policy results

Branch `feat/fuel-clearing-step1`

September 24, 2026

Abstract

AeroMAPS is currently *told* the share of each fuel and computes the resulting cost. This work inverts that: a convex optimisation programme is given a policy — an obligation, a penalty, growth limits, capacities — and returns both the volume of each fuel produced and the price at which it clears. The programme is solved once over all regions, all pathways and all prospective years, so it anticipates: it builds ahead of an obligation it can see coming.

This report states the model in full, describes every parameter, and separates two classes of result that are easy to confuse: those obtained from the optimisation kernel *alone*, at a demand fixed from outside, and those obtained with the kernel *chained into AeroMAPS*, where the price it sets feeds airline costs, airfares and therefore back into demand. It reports what was verified, what was found to be broken along the way, and what the mode cannot yet do.

The headline results: with scarcity switched off the new mode reproduces the existing one to 1.9×10^{-9} across the whole chain; with scarcity on it produces a quantity AeroMAPS has never had — a scarcity rent.

That rent turned out to be exactly what made the coupled fixed point hard to reach, and chasing it down is the other half of this report. The market's price is a Lagrange multiplier; a multiplier is a subgradient of a piecewise-smooth value function; and where that function kinks the price is an *interval* rather than a number, which no fixed-point iteration can be asked to converge on. Giving the energy balance a demand slope — quantity from supply, price from demand, as in any capacity-constrained market — makes the price single-valued again, and is constructed so that it is exactly inert once the loop settles. The coupled model now converges at full marginal pricing, in 60 sweeps and with no acceleration, to a solution of the unmodified programme.

A five-pathway bench then shows that how a supply limit is shaped matters only where a limited fuel is the marginal one, and that where every fuel is at its limit the price comes from demand. The report closes with the decisions framing step 2 — one optimisation, perfect foresight, regions interacting through one global pool — and its first two steps: the pool's volume side in the kernel, and flows between regions carried through AeroMAPS with an explicit matrix standing in for the market.

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1 What this is, and what it answers

The existing AeroMAPS energy chain answers: *given these fuel shares, what does the scenario cost?* It cannot answer *given this policy, what fuel gets made?*, because the quantity produced in response to a policy is exactly what a market determines and the current model is handed it as an input.

The fuel market adds a second mode, opt-in and default off. Both modes coexist; they answer different questions. The new one takes an obligation, a penalty for missing it, a production cost and a set of physical limits per pathway, and returns:

- **volumes** — how much of each fuel is produced, in each region, each year;
- **prices** — what the energy is worth at the margin, what the obligation costs to comply with, and what a growth limit is worth relaxing.

The prices are not assumed or calibrated. They are the Lagrange multipliers of the programme's constraints, which is what makes convexity worth paying for: a convex programme has one optimum, and its multipliers are certified prices that do not depend on where the solver started.

2 The model

2.1 The programme

Over regions $r \in \{1..R\}$, pathways $p \in \{1..P\}$ and prospective years $t \in \{1..T\}$, with decision variables $q_{r,p,t} \geq 0$ (energy produced, MJ) and $x_{r,t} \geq 0$ (obligation released by paying the penalty, MJ):

$$\min_{q, x} \sum_t \delta_t \left[\underbrace{\sum_{r,p} \left(c_{r,p,t} q_{r,p,t} + \frac{c_{r,p,t} \gamma_{r,p} K_{r,p,t}}{n+1} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^{n+1} \right)}_{\text{production cost, integral of the marginal cost}} + \sum_r B_{r,t} x_{r,t} \right] \quad (1)$$

subject to, for every region r and year t :

$$\sum_p q_{r,p,t} = D_{r,t} \quad \text{energy balance} \quad \rightarrow \lambda_{r,t}^E \quad (2)$$

$$\sum_{p \in \mathcal{S}_r} q_{r,p,t} + x_{r,t} \geq m_{r,t} D_{r,t} \quad \text{obligation} \quad \rightarrow \lambda_{r,t}^M \quad (3)$$

$$\sum_{p \in \mathcal{N}_r} q_{r,p,t} + x_{r,t}^{\text{sub}} \geq \mu_{r,t} D_{r,t} \quad \text{sub-obligation} \quad \rightarrow \lambda_{r,t}^S \quad (4)$$

$$q_{r,p,t} \leq s_{r,p} + (1 + g_{r,p}) q_{r,p,t-1} \quad \text{growth limit} \quad \rightarrow \lambda_{r,p,t}^R \quad (5)$$

with $q_{r,p,0} \equiv q_{r,p}^{\text{init}}$ given. $\mathcal{S}_r$ is the set of pathways *eligible* for region r 's obligation and $\mathcal{N}_r \subseteq \mathcal{S}_r$ the narrower set for its sub-obligation.

Three properties of this programme carry the design.

One solve, not one per year. Constraint (5) ties t to $t - 1$, so solving year by year would discard anticipation. Sustainable output above the obligation in early years is a *result*, not an artefact: with perfect foresight the cheapest way to meet a step is to be ready before it arrives.

Convex, so the multipliers are prices. λ^E is the marginal cost of energy; λ^M is the compliance price. A fixed point found by iteration would not come with that guarantee.

The growth limit is a sum, not a maximum (§2.6).

2.2 Lagrange multipliers, and the exact sense in which they are prices

The claim in the box above — “ λ^E is the marginal cost of energy” — is the load-bearing one in this whole design, and it is worth deriving rather than asserting, because the same derivation shows precisely where it stops holding.

The Lagrangian. Write the objective (1) as $C(q, x)$ and attach a multiplier to each constraint, signed so that every multiplier is non-negative on an inequality:

$$\begin{aligned} \mathcal{L}(q, x; \lambda) = & C(q, x) + \sum_{r,t} \lambda_{r,t}^E (D_{r,t} - \sum_p q_{r,p,t}) \\ & + \sum_{r,t} \lambda_{r,t}^M (m_{r,t} D_{r,t} - \sum_{p \in \mathcal{S}_r} q_{r,p,t} - x_{r,t}) \\ & + \sum_{r,p,t} \lambda_{r,p,t}^R (q_{r,p,t} - s_{r,p} - (1 + g_{r,p}) q_{r,p,t-1}). \end{aligned} \quad (6)$$

Stationarity is a statement about margins. Setting $\partial \mathcal{L} / \partial q_{r,p,t} = 0$ for a pathway produced in strictly positive quantity gives, after dividing through by the discount factor δ_t :

$$\underbrace{c_{r,p,t} \left(1 + \gamma_{r,p} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^n \right)}_{\text{marginal production cost}} + \underbrace{\lambda_{r,p,t}^R - (1 + g_{r,p}) \frac{\delta_{t+1}}{\delta_t} \lambda_{r,p,t+1}^R}_{\text{scarcity of the right to grow}} = \lambda_{r,t}^E + \lambda_{r,t}^M \mathbf{1}[p \in \mathcal{S}_r]. \quad (7)$$

Read from the right: what a unit of this fuel is *worth* to the system — the energy it supplies, plus the obligation it discharges if it is eligible — equals what it costs at the margin, plus the shadow value of the growth room it consumes. That is an identity, not a modelling assumption. Three things follow from it directly.

- **Equation (7) is why the ramp-up multiplier never appears in any published price.** It is already inside λ^E and λ^M . A model that added λ^R to the price the airline pays would count it twice.

- **A pathway not produced** has $\partial\mathcal{L}/\partial q \geq 0$ instead: it is worth less than it costs. The programme’s choice of which fuels to build is exactly this comparison.
- **The term $-(1+g)\lambda_{t+1}^R$ is anticipation, written down.** Growth room used this year buys growth room next year. Its sign is negative: building now is *worth* something later. This single term is why the solve builds ahead of an obligation (§5), and it exists only because all years are solved at once.

Complementary slackness. For every inequality, $\lambda \geq 0$, the slack is ≥ 0 , and their product is zero:

$$\lambda_{r,t}^M \cdot \left(\sum_{p \in \mathcal{S}_r} q_{r,p,t} + x_{r,t} - m_{r,t} D_{r,t} \right) = 0. \quad (8)$$

An obligation that is over-met has *no* compliance price; one that is exactly met has one. This is a testable property rather than an interpretation, and it is what the parametrised test over seven regimes checks (§4). Every degeneracy found in this work was found by it failing.

The envelope theorem is the actual claim. Let $V(D)$ be the optimal value of the programme as a function of the demand vector. Then at an optimum

$$\lambda_{r,t}^E = \frac{\partial V}{\partial D_{r,t}}. \quad (9)$$

The multiplier is the derivative of the cost of the whole system with respect to asking it for one more MJ — every knock-on effect included, across pathways, across years, through the ramp-up chain. *That* is what makes it a price and not a bookkeeping quantity, and it is unavailable from any allocation rule: a rule can report what each fuel cost, but only an optimum can report what the next unit would cost.

Why convexity is not a technicality. Equation (9) needs strong duality, which for this programme follows from Slater’s condition (the feasible set has a relative interior whenever the obligation is reachable, and the buy-out x guarantees that it is). Convexity also delivers what no iterative scheme can: the optimum is global and the duals are certified, so the answer does not depend on where the solver started. This is the entire reason the market is a dedicated convex solve rather than another block inside the MDA (§3.3).

The caveat that the rest of this report is about. Equation (9) is an equality only where V is differentiable. In general the correct statement is

$$\lambda \in \partial V(D) \quad (\text{the subdifferential}),$$

and V is piecewise smooth, not smooth: it has a kink wherever the set of tight constraints changes. At such a point ∂V is an *interval*, every member of it satisfies the KKT conditions, and the solver returns one of them with no way to prefer another. The price is genuinely not determined by the cost side there.

Nothing is wrong with the optimisation. But a quantity that is set-valued cannot be the input to a fixed-point iteration, and that — not stiffness, not scaling, not the solver — is why the coupled loop failed at $w = 1$ (§6.4), and what §6.5 does about it.

2.3 Every parameter

2.4 The prices the model produces

The *marginal price* of a pathway aggregates the constraints one unit of it relaxes:

$$\text{MC}_{r,p,t} = \lambda_{r,t}^E + \lambda_{r,t}^M \mathbf{1}[p \in \mathcal{S}_r] + \lambda_{r,t}^S \mathbf{1}[p \in \mathcal{N}_r]. \quad (10)$$

A sub-mandated unit carries both obligation multipliers. That is not double counting: it genuinely relaxes two distinct constraints.

2.5 What the airline actually pays

This section answers a question the earlier write-ups left implicit.

Two prices exist for each pathway and they are not the same. The **average cost** is what the fuel produced actually cost, per unit:

$$AC_{r,p,t} = c_{r,p,t} \left(1 + \frac{\gamma_{r,p}}{n+1} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^n \right), \quad (11)$$

against the marginal price (10). The gap between them is **inframarginal rent** — profit earned on every unit produced more cheaply than the last one. The pricing weight interpolates:

$$\text{mfsp}_{r,p,t}^{\text{mkt}} = (1-w) AC_{r,p,t} + w MC_{r,p,t} - \underbrace{\left(\text{net}_{r,p,t} - \text{gross}_{r,p,t} \right)}_{\text{tax wedge, §3.4}} \quad (12)$$

and AeroMAPS’s existing machinery then weights that across pathways into the one number the airline sees, $\text{type_mean_mfsp} = \sum_p \text{share}_p \cdot \text{mfsp}_p^{\text{mkt}}$, which drives operating cost, airfare and demand.

So: fuel cost, or compliance, or both? The honest answer is *both, always* — but not by addition, and this is the easiest thing in the model to get wrong.

Region A, 2050	Airline pays	λ^M	SAF share
No obligation at all	0.01200	0	0 %
70 % obligation, $w = 0$	0.01982	0.01117	70 %
70 % obligation, $w = 1$	0.01982	0.01117	70 %
+ scarcity, $w = 0$	0.02115	0.07917	70 %
+ scarcity, $w = 1$	0.06742	0.07917	70 %

The obligation is paid for at $w = 0$: the bill goes from 0.01200 to 0.01982, a **65 % increase**. It is paid through the *mix* — the obligation forces 70 % of the fuel to be the expensive kind — not through a separate charge. The cost of compliance is *embedded*, not itemised.

λ^M is not added on top of that, and must not be. It is the *marginal* cost of the obligation — what one more unit of it would cost. The average cost increment is $0.01982 - 0.01200 = 0.00782$, which is exactly $0.70 \times \lambda^M$: share times marginal cost. Adding λ^M to the delivered price would give 0.03099, a 158 % increase, **charging the obligation twice**.

What w changes is average versus marginal, not paid versus unpaid. At $w = 1$ every unit is priced at the *last* unit’s cost. With nothing scarce that is the same number (rows 2 and 3 are identical — marginal equals average when the supply curve is flat). With scarcity it is not: 0.06742 against 0.02115, and the difference is inframarginal rent.

It is never “compliance only”. λ^M is a premium on top of the energy price, never a substitute for it.

The carbon tax arrives separately, as its own `DirectOperatingCosts` line — which is exactly why (12) subtracts the tax wedge back out (§3.4). Same double-counting logic, different instrument.

“Why average across pathways at all? Should there not be *one* market price — or one per obligation, one for bio and one for e-SAF?”

At $w = 1$ there already is, and the weighted average is an identity rather than a modelling choice. Under uniform pricing every eligible producer is paid the same, so (10) takes one value per *eligibility class*, not one per pathway. Measured on a five-pathway case with a 70 % obligation and a 25 % e-SAF sub-target:

	$w = 0$	$w = 0.5$	$w = 1$
distinct per-pathway prices	5	5	3
share-weighted average	0.041184	0.042087	0.042990
$\lambda^E + m\lambda^M + m_s\lambda^S$	0.042990	0.042990	0.042990
gap	-0.001806	-0.000903	8×10^{-13}

Five pathways, **three** prices: 0.012 for fossil, 0.0322 for “counts towards the broad obligation”, 0.0996 for “counts towards both”. One per obligation, plus the energy price — exactly the structure a certificate market has. `hefa_fog` is paid 0.0322 though it cost 0.02317; the 0.00903 difference is its rent.

λ^K and λ^R are not in it, and must not be added: they are already *inside* the price, on the producer’s side. For each fuel, price = cost + λ^K + λ^R , net of next year’s growth-limit rent (§5.2): they say how the payment splits between cost and rent. Adding them would charge the rent twice.

So λ^E , λ^M and λ^S are the market prices. `{p}_market_mfsp` is those three numbers re-expressed per pathway so that AeroMAPS’s existing per-pathway grammar can carry them, and the weighted average is the arithmetic that reassembles them. Writing $\lambda^E + \sum_o m_o \lambda^o$ directly would give the same number and would say more clearly what it is.

Below $w = 1$ the average stops being an identity and starts being an assumption, and the gap in the table is exactly w times the rent per unit — at $w = 0$, $-0.001806 = 0.00903 \times 0.20$, `hefa_fog`’s rent times its share. So w is not “average versus marginal” in the abstract. It is **the share of scarcity rent the airline is charged**, and the weighted average is how that dial is implemented.

One gap this exposes. λ^S is computed and never published: the discipline emits `fuel_market_energy_price` and `fuel_market_compliance_price` and no sub-mandate price, and it does not wire `submandate_share` through from the scenario at all — the sub-mandate lives in the kernel and is exercised only in §8.2’s kernel-level case. A price vector that is genuinely “one per obligation per region” needs the discipline to carry an arbitrary number of obligations and publish a dual for each. It carries one.

One thing the airline does *not* pay, and arguably should. The buy-out payment Bx appears in objective (1) but in no price output. If the market releases part of the obligation, that penalty is borne by nobody in the cost chain. At $w = 1$ the *rate* is signalled correctly (λ^M is capped at B), but the *payment* is not accounted anywhere. This is a known gap, not a modelling choice.

A useful consistency check falls out: with no scarcity at all, $\lambda^E = c_k$ (kerosene) and $\lambda^M = c_s - c_k$, so a sustainable pathway’s marginal price is exactly c_s — which is also its average cost. The two endpoints of (12) coincide and w is inert. **w only matters where something is scarce**, from *either* the capacity term or a binding growth limit.

2.6 Two choices that keep the problem convex

Capacity squeezes, it does not wall. Production is not capped. Instead the marginal cost rises with utilisation, $c(1 + \gamma(q/K)^n)$, whose integral appears in (1). A hard cap would make the price jump discontinuously at $q = K$; since that price feeds airfares, which feed demand, which feeds fuel demand, a jumping price makes the coupled loop oscillate instead of settle. Figure 1 shows the shape and the rent.

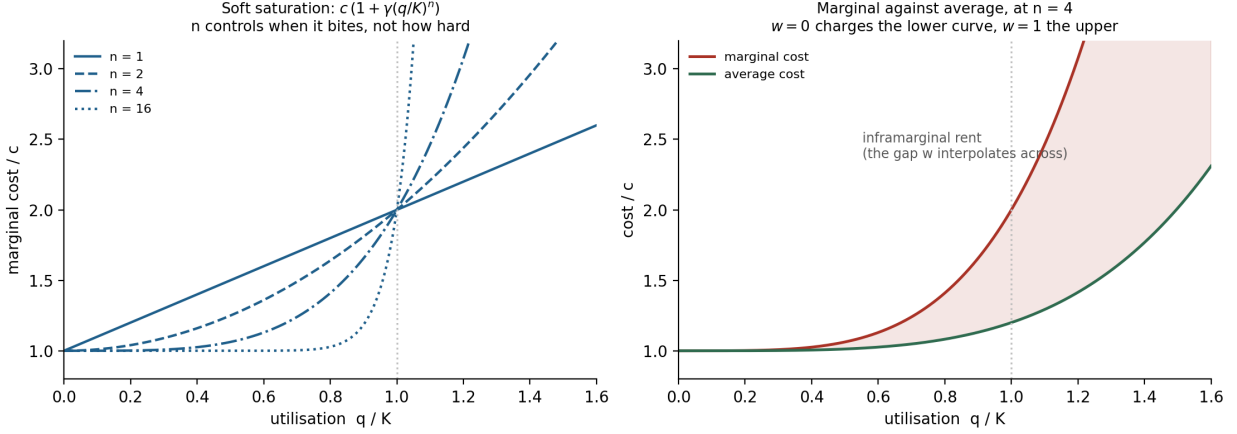


Figure 1: Left: n decides *when* the squeeze bites, not how hard. Note the consequence — below $q = K$ a *larger* n gives a *smaller* markup, which is why stiffness reduces the price on a bench running under capacity. Right: the gap between marginal and average cost is the rent that w interpolates across.

The growth limit has to be a sum. AeroMAPS's optimisation mode limits growth by Eq. (12) of the published work — the *more permissive* of a relative rate and a fixed volume addition:

$$E_t \leq \max\{(1 + \tau)E_{t-1}, E_{t-1} + \Delta E\}. \quad (13)$$

The region under a maximum of two affine functions is not a convex set (Figure 2, right), so (13) cannot be handed to a convex solver. This is not a tolerance problem; the feasible set is the wrong shape. Constraint (5) replaces the maximum with a *sum*, which is its convex outer relaxation: it admits everything (13) admits, plus a margin **measured at 0.73 %, independent of g** .

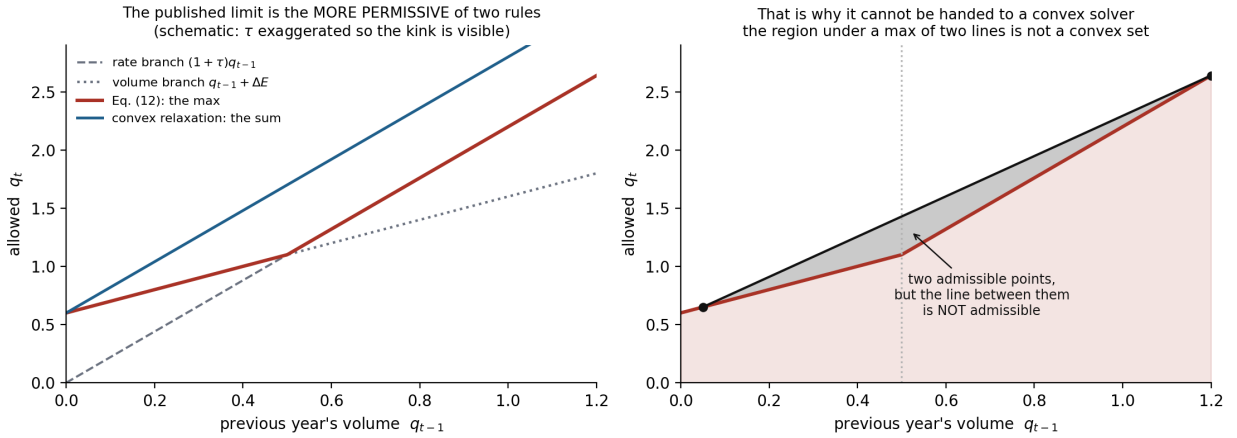


Figure 2: Why the published growth limit cannot go into the programme, and what replaces it. Right: two admissible points whose connecting line is inadmissible — the definition of a non-convex set.

The seed s in (5) is Eq. (13)’s volume branch ΔE , so it is *not* a free parameter: the published calibration (0.2 EJ/yr at an ASK share of 0.1549) puts it at about **1.5 % of annual demand**. It matters — on the bench the seed moves the compliance price by a factor 4.3 across a plausible range.

3 How it plugs into AeroMAPS

3.1 Two levels of result, and why the distinction matters

This section answers the second question the earlier write-ups left implicit.

Results in this report come from one of two set-ups, and conflating them would be a serious misreading.

Why the policy cases are kernel-only, deliberately. Chaining changes no *ordering* at $w = 0$, while costing minutes per case. (At $w > 0$ it also used to introduce a convergence problem, §6.4; that is now resolved, §6.5, but the cost per case remains.) The kernel-level results are therefore the *policy* answer.

How much the chain damps a price effect is NOT measured, and an earlier version of this box said it was — “roughly a tenth”, cited to §6.3. That section says something different and correct: the *traffic* response is about a tenth of the *price* response. The two grids cannot be differenced to recover the damping, because they do not share a baseline — the kernel grid sweeps γ at a fixed growth limit of 0.30, while the coupled grid compares *no scarcity at all* against a growth limit of 0.20 with a capacity and a γ . So every magnitude in §8 should be read as kernel-level with an *unquantified* coupled damping, which at $w = 1$ is where it is most likely to matter.

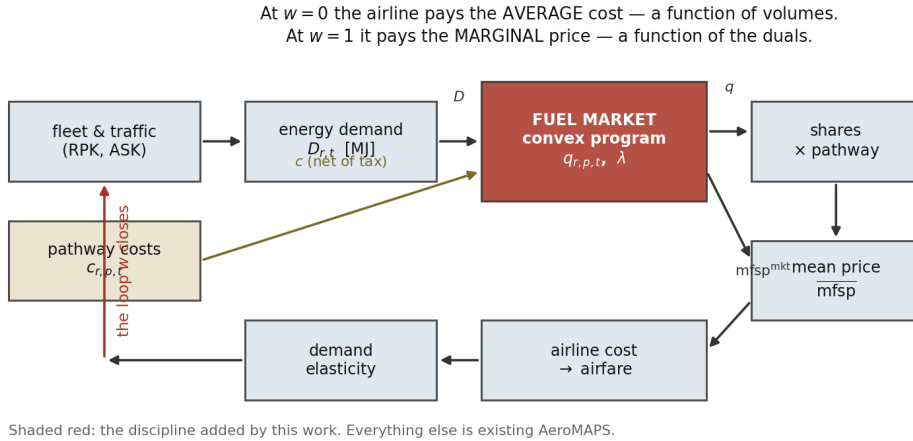


Figure 3: Where the market sits. Shaded: the discipline added by this work. The loop on the left is closed only in coupled mode, and only carries a discontinuous quantity when $w > 0$.

3.2 What the discipline replaces

In market mode the existing allocator `EnergyUseChoice` is **not instantiated**: the market decides the volumes rather than being handed shares. It emits the same nine output families under the same names, computed by a *shared* function extracted from the allocator rather than a second copy — two copies would drift, and the failure is silent, since a share family disagreeing with its volumes rescales CO₂ with no error raised.

The mode is one configuration key, default off, propagated to each region as a process-level keyword. It is refused outside the unified-MDA execution mode, and three guards raise at set-up rather than producing plausible-looking numbers: drop-in fuels only, top-down cost model only, exactly one residual pathway.

3.3 The MDA, and why it is an awkward instrument for this particular coupling

AeroMAPS closes its loops with a Multi-Disciplinary Analysis: GEMSEO orders the disciplines, identifies the strongly coupled block (here 63 variables across two regions), and sweeps it with Gauss–Seidel until a residual norm falls below tolerance. For the couplings AeroMAPS was built around — fleet renewal, energy intensity, traffic — this is exactly right. The market is a worse fit, and it is worth being explicit about why, because the limitation is structural rather than a matter of settings.

An MDA is a fixed-point iteration, and it asks for more than it looks like. Gauss–Seidel converges when the composite map from coupling variables to coupling variables is a contraction near the solution. In this loop that map is

$$p \mapsto D(p) \mapsto \Lambda(D(p)) = p',$$

where $D(\cdot)$ is the whole fleet–cost–airfare–elasticity chain and Λ is the market’s price. Convergence therefore requires Λ to be not merely bounded but *Lipschitz* in demand, with a small enough constant. At $w = 0$ it is: the price is the average cost, an explicit smooth function of the volumes. At $w = 1$ it is a multiplier, and §2.2 established that a multiplier is a subgradient of a piecewise-smooth value function. Its Lipschitz constant is infinite at every kink. No choice of relaxation factor or acceleration method repairs that, because the failure is in the map, not in the scheme applied to it — a point this work initially got wrong and had to retract (§9).

The disciplines exchange point values, and a set-valued quantity has none. The MDA’s contract is that each discipline is a function: same inputs, same outputs. At a kink the market is a *correspondence* — the KKT system admits an interval of prices, and which one comes back is a property of the solver’s trajectory. It is consistent with the maths and inconsistent with the contract. Nothing in GEMSEO can detect this; the discipline looks like it is returning a number, and it is.

The convergence test measures the wrong thing. The residual is a norm over all 63 coupling variables, in their own units. A market that has settled to eleven digits can sit inside a residual dominated by `region_B:ask` at 10^{12} , and a market that is flipping between two active sets can hide inside a residual that is falling for unrelated reasons. Diagnosing §6.4 required instrumenting the discipline itself — and four successive instrumentation attempts were wrong before one was right (§9), each producing a confident, false reading.

The market is solved many times to answer one question. The convex programme is solved *in full* — all regions, pathways and years, with perfect foresight — at every sweep. A converged coupled case costs 23 solves at $w = 0$ and 60 at $w = 1$; a failing one costs 202. Each is a globally optimal answer to a question that is about to be superseded.

What the alternative would be. All of the above points the same way: the demand response belongs *inside* the programme rather than outside it. A single optimisation maximising surplus net of cost, with demand as a decision variable, has no outer loop, no set-valued exchange and

no residual to interpret — its optimality conditions simply include the demand relation. That is a large change to how AeroMAPS is organised and it is not proposed here.

What *is* done here (§6.5) is the smallest useful step in that direction: a local linearisation of the demand curve is passed into the programme, enough to make the price single-valued, constructed so that it is exactly inert once the loop settles. The MDA stays; it is simply no longer asked to iterate on a quantity that has no value to iterate on.

3.4 Decide on net, report on gross

The market chooses using `net_mfsp` — cost *including* carbon tax and subsidies — because that is what determines which pathway is marginal. This is not cosmetic: on the bench a carbon tax of only 5 EUR/t already narrows the sustainable premium by 3 %, and extrapolating the emission factors the premium reaches zero near **164 EUR/t**, where the ordering reverses and the market buys sustainable fuel unprompted. A market clearing on gross cost could never produce that.

But `DirectOperatingCosts` adds the carbon tax as its own line. Publishing a net-basis price would count the tax twice — hence the wedge subtracted in (12). **Decide on net, report on gross, and the tax is counted once.**

A *passenger*-level tax is deliberately excluded from the objective: it is uniform across fuels, so it cannot change which pathway is marginal. It still reaches the market through demand, which is the correct channel.

4 Verification

An analytic case pins the conventions. One region, one year, saturation known in closed form, so both prices are known exactly. This caught a sign error that documentation could not have: `cvxpy` reports the equality’s multiplier negated, so the energy price came out as -0.012 against an expected $+0.012$ — a plausible number with the wrong sign.

98 tests on the fuel market — 76 on the kernel, 8 on the discipline, 14 on the flow plumbing of §13 — covering the balance, the obligation, the growth limit, the capacity ceiling, the pool, scale invariance, continuity and input hygiene. The full suite passes.

Complementary slackness, across seven regimes. A multiplier may be non-zero only where its constraint is tight. This single property is what all three real defects (§9) violated, and nothing tested it. It is now parametrised over the *edges* — zero obligation, full obligation, a constraint that cannot bind — because a mid-range case sees none of them. Worst residual: 1.4×10^{-8} , solver noise; the phantom compliance price scored 0.46 on the same measure.

5 Results without the chain (kernel only)

5.1 It reproduces the current allocator

Given the reference run’s shares, with saturation off and the growth limit loose, the market lands on the reference allocation with a residual below 10^{-8} of demand — an order of magnitude under solver noise.

5.2 The build-ahead: right motive, wrong instrument

At $g = 15\%/yr$ against the ReFuelEU steps, region A holds **more** sustainable fuel than it owes in every year between steps: 14.1 % in 2034 against a 6 % obligation, 28.9 % in 2039 against 20 %. The growth limit cannot take it from 6 % to 20 % in the year the step lands, so it starts early, and it overshoots by the *minimum* that clears the next step, discounting penalising early spend.

The motive is real, and it is penalty avoidance. The obligation (3) is written per year and there is no banking, so a sustainable MJ produced in 2045 has no value *as fuel* in 2046. But the growth limit ties what can be produced next year to what is produced this year. Producing early therefore has a value: it relaxes every later growth limit, and that relaxation is worth the discounted compliance cost — up to the penalty — it saves when the step lands. The price identity shows it directly: for a producing pathway,

$$MC_{p,t} = c_{p,t} + \lambda_{p,t}^K + \lambda_{p,t}^R - \frac{(1+g)\lambda_{p,t+1}^R}{1+\rho},$$

and the last term is exactly what producing now is worth to next year’s allowance — the anticipation, priced. A buyer with perfect foresight who knows a step is coming *should* pay to be able to meet it.

Measured. Sustainable share against the obligation, five-pathway kernel case, hard capacities, varying only the growth limit:

	2039	2040–44	2045–48	2049	2050	max over-compliance
obligation	0.20	0.34	0.42	0.42	0.70	—
$g = 1000$ (slack)	0.200	0.340	0.420	0.420	0.700	0.0000
$g = 0.30$	0.060	0.138→0.340	0.420	0.504	0.700	0.0838
$g = 0.15$	0.060	0.129→0.340	0.420	0.557	0.700	0.1365

With a slack growth limit the build-ahead is exactly zero: without a limit there is no future penalty to anticipate. With one, the over-compliance sits just before a step (2049, before 70 %), where it buys the most relief, and the same rows *under-comply* in 2039–41, where no anticipation could have ramped fast enough and the penalty is simply paid.

What is wrong is the instrument. Because the growth limit is written on *production*, the only way the programme can say “be able to meet 2050” is to make the fuel in 2049 and burn it. In reality the same anticipation takes the form of plants built ahead of the step, or of airlines signing offtake contracts early so that the plants reach financing — neither of which burns 8 to 14 extra points of the mix a year early. The *volume* of early sustainable fuel is therefore overstated, with the early fuel bill and early CO₂ savings; the *existence* of anticipation is not. The fix is a capacity variable (§11).

Corrected. Earlier versions of this section first called the build-ahead “what a firm reading a published mandate would do”, then withdrew that on the grounds that nothing in the programme rewards buying early. The second statement was too strong: the growth limit is exactly such a reward.

Perfect foresight is kept (§11). The case against it — one planner’s programme standing in for separate agents — reduces to three assumptions, stated rather than changed: there is no banking of compliance; one discount rate prices the airline’s fuel bill and the producer’s plant alike (the capacity variable addresses this by annuitising capex at a producer financing rate); every announced obligation is taken as fully credible.

The penalty is needed exactly once, at the 2035 step and only at the tightest growth limit. The compliance price is **zero in every year where the market is ahead**.

5.3 Which parameters move the answer

Two readings that a single-parameter view gets wrong. γ ’s **1.7 % is misleading**: it was measured at one stiffness, and at $n = 1$ the same range moves the uplift 4 %→16 %. The two saturation

parameters are not separable. And γ 's real effect at this baseline is on the *compliance* price, which it moves +57 % — scarcity raises the cost of the obligation well before it raises the average fuel bill.

The penalty is inert until it is cheap enough to be worth paying. The obligation's own marginal cost here is 0.0546 EUR/MJ. Set B above that and it never binds; set it below and the market stops building and pays instead, and the *delivered* price falls 8.3 %. A penalty priced under the cost of compliance is a subsidy to non-compliance.

5.4 Capacity sized on the obligation is undersized

With a binding growth limit, anticipation pushes utilisation to $q/K = 2.01$ between steps: the market builds twice the plant the obligation asks for. Saturation disciplines it — at $\gamma = 1$ the peak falls to 1.30.

6 Results with the chain (coupled)

6.1 End-to-end reproduction

With nothing scarce, the coupled mode must reproduce the existing model. It does, and not only in the volumes:

An independent confirmation fell out: the compliance price came back at 0.01083, which is the *net*-basis premium (0.010830) and not the gross one (0.011170). The decide-on-net rule is doing what it claims.

6.2 Scarcity reaches the traffic loop

Expensive fuel makes tickets dearer and some journeys do not happen. AeroMAPS could not previously produce that chain of consequences *from a fuel policy*, because nothing in it priced scarcity.

6.3 The coupled saturation grid

Swept through the full MDA, so volumes respond. Delivered price % / RPK % against the no-scarcity run:

The anchored run agrees with the rigid one wherever the rigid one worked. This is the system-level check that §6.5's inertness argument survives contact with the full chain, and it is worth more than the kernel test because it exercises every discipline between the market and the demand:

- all 13 $w = 0$ cells: agree to $\sim 10^{-10}$ relative — unchanged, as required;
- the 3 $w = 1$ cells that converged before: agree to 3×10^{-8} ;
- 6 $w = 1$ cells that previously failed: now converge.

The demand slope changed nothing that already worked, and fixed what did not.

Two results survive the feedback intact. The kernel-level *ordering* holds — uplift still falls monotonically in n , at $w = 1$ as well now that the row is legible. And **the traffic response is consistently about a tenth of the price response**: +42.1 % on price gives −4.7 % on RPK. That ratio is the product of two independently measured transmissions (fuel is 13.1 % of the airfare; demand elasticity 0.888). Rule of thumb for this bench: **a 10 % fuel-price rise costs about 1.2 % of traffic.**

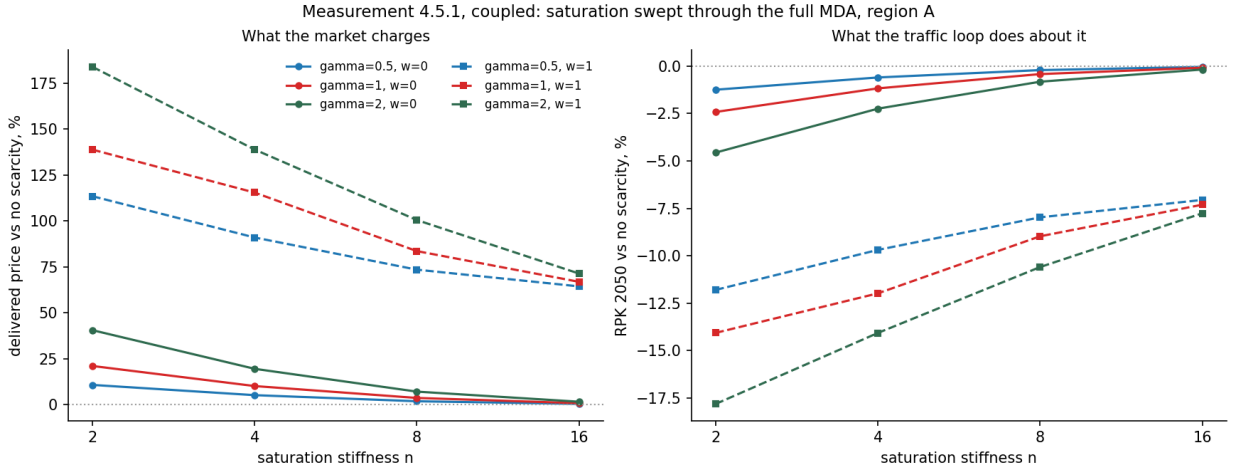


Figure 4: The same grid drawn. Gaps are cells that did not converge; they are not interpolated over.

6.4 The coupled fixed point, and why it failed at $w = 1$

At $w = 1$ the loop does not converge under AeroMAPS’s default settings: the residual stalls at 8.6×10^{-2} after 200 sweeps. Damping helps but does not close it, and acceleration — which an earlier version of this report credited with fixing it — does not (§9).

The mechanism was first *inferred* from the shape of the residual. It has since been *measured*. The kernel now reports an `active_signature`: a digest of which constraints are tight, taken from the primal slacks and never from the multipliers, since whether a multiplier is non-zero is precisely the question that degenerates. Two sweeps with the same signature solved the same linear system for their prices. The measurement (`fuel_clearing_step1/convergence.py`, Fig. 5) is Table 5.

The two sets it alternates between differ in exactly two cells:

$$\{\text{obligation tight : 31, growth limit tight : 31}\} \longleftrightarrow \{\text{obligation tight : 29, growth limit tight : 33}\}.$$

The mechanism, precisely. In two (region, year) cells the binding constraint trades places between *the obligation binds* and *the growth limit binds*. On one side the obligation is met by volume and λ^M is the cost of substituting one fuel for another, about 0.012 EUR/MJ. On the other the obligation has become physically unreachable and the only way to discharge it is to pay — so λ^M snaps to the buy-out, 0.30 EUR/MJ. The delivered price jumps 23–29 % when it does.

The volume is identical on both sides. It is pinned by the growth limit. Only the price differs. That is not a solver artefact: at the crossing point the value function has a genuine kink, relaxing the growth limit saves nothing (the obligation is already met) while tightening it costs the buy-out, so the right and left derivatives differ and ∂V is the whole interval $[0.012, 0.30]$ — a factor of 25. This is exactly the caveat of §2.2, met in practice.

So w decides whether the model couples to a well-defined function of its own state, not how volatile the price is. At $w = 0$ the delivered price is the average cost: a function of volumes, which move by 1–2 %. At $w = 1$ it contains λ^M , and at a kink λ^M is not a function of the state at all.

Two explanations were tested and **rejected**. The bad years sit next to a ReFuelEU step, so the staircase obligation looked responsible — but replacing it with a smooth ramp through the same points still fails: smoothing moves the boundary rather than removing it. And the measured loop gain, ≈ 0.32 , is far below 1; a gain that low cannot diverge, which is what ruled out “the prices are volatile” at the outset.

6.5 The fix: quantity from supply, price from demand

The diagnosis dictates the remedy. Where the supply curve is vertical — a growth limit exhausted, an obligation reachable only by buying out — the quantity is determined and the price is not. That is not a pathology; it is the ordinary situation in any capacity-constrained market, and every such market resolves it the same way: **the quantity comes from supply and the price comes from demand**. The programme was solving a supply curve at a fixed quantity and then being asked for a price the supply side does not contain.

So the balance is given a slope. Around the incoming demand $D_{r,t}$ and the price $p_{r,t}^0$ that demand was formed at, introduce $a_{r,t}$ (free) and replace (2) and (3) with

$$\sum_p q_{r,p,t} = D_{r,t} + a_{r,t}, \quad \sum_{p \in \mathcal{S}_r} q_{r,p,t} + x_{r,t} \geq m_{r,t} (D_{r,t} + a_{r,t}), \quad (14)$$

adding to the objective the consumer surplus given up, to second order:

$$\sum_{r,t} \delta_t \left[-p_{r,t}^0 a_{r,t} + \frac{a_{r,t}^2}{2\beta_{r,t}} \right], \quad \beta_{r,t} = \eta \frac{D_{r,t}}{p_{r,t}^0} \geq 0. \quad (15)$$

The quadratic keeps the programme convex. Stationarity in $a_{r,t}$ then reads

$$\boxed{\lambda_{r,t}^E + m_{r,t} \lambda_{r,t}^M = p_{r,t}^0 - \frac{a_{r,t}}{\beta_{r,t}}} \quad (16)$$

— the delivered marginal price equals the inverse demand at the quantity served. The left-hand side is what the airline faces per MJ: the energy, plus the share m of obligation that MJ drags with it (§2.5). The identity holds to 5×10^{-11} in the kernel tests, and it is the entire mechanism. On a vertical stretch of supply, a moves until the price sits where demand wants it; the gap that opens between that price and the marginal production cost is a **scarcity rent**, which is what a capacity-constrained producer earns, not an error.

This is exact, not a relaxation. At a fixed point of the coupling loop the price the market returns *is* the price the demand was formed at, so $p^0 = \lambda^E + m\lambda^M$, hence $a = 0$ by (16), hence (14)–(15) reduce to (2)–(3) identically. The loop therefore converges to a solution of the *unregularised* programme or it does not converge at all. There is no fixed point at which the added term is still bending the answer. Measured: $|a|/D = 6 \times 10^{-11}$ at convergence.

And η does not have to be right. It sets how far the price moves per sweep and nothing else. Linearising the loop gives the requirement $\beta > \frac{1}{2}|dD/dp|$: over-stating the demand response is safe and merely slow, under-stating it by more than a factor of two overshoots.

Two further consequences. First, a **negative result worth keeping**: anchoring the *volumes* instead — a proximal term $\frac{\rho}{2}\|q - \hat{q}\|^2$ — does not work, and cannot, because the primal is identical on both sides of the kink. It moves λ^M across 0.6% of the interval it is free in. It is retained as an optional, separately weighted device for degeneracies in the *primal*, off by default.

Second, all of this runs on **plain Gauss–Seidel with no acceleration**. The acceleration that was previously thought to help is gone from the coupled grid along with the claim.

“Away from a kink the supply curve *does* have a price at the quantity handed in. Is it not bizarre to take the price from demand there — and might the answer not be the true crossing of supply and demand?”

The objection is right and the title of this section is what invites it. Away from a kink the price is *not* taken from demand, in the following precise sense.

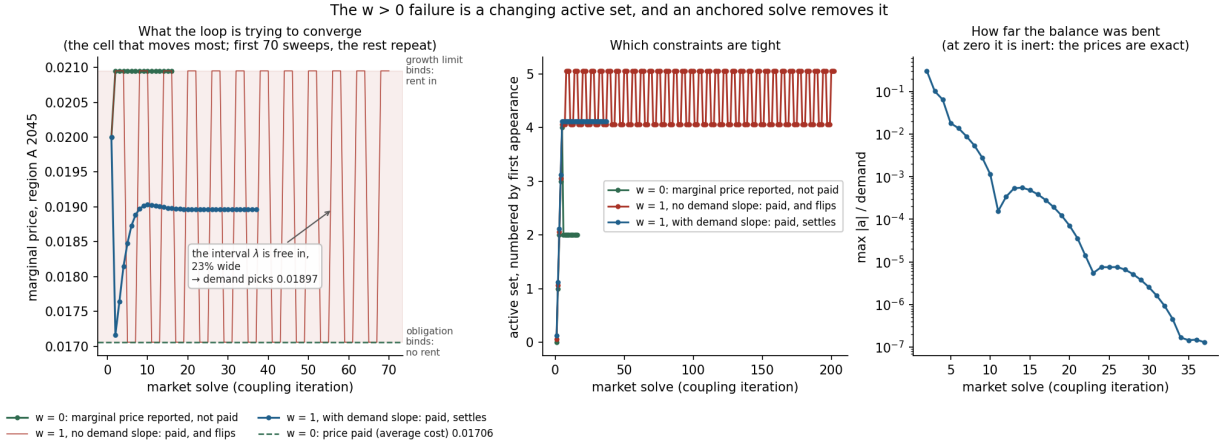


Figure 5: **Left**, region A 2045. The shaded band is every price the solver may return at the kink: at its bottom (0.0171) the obligation binds and there is no rent; at its top (0.0210) the growth limit binds and its rent is included. Green, $w = 0$: airlines pay the *average* cost, 0.0171 (dashed); the marginal price drawn is reported but paid by nobody, so the solver’s pick never reaches traffic and the loop settles in 16 solves. Red, $w = 1$ without a demand slope: the paid price jumps between the band’s ends forever. Blue, $w = 1$ with a demand slope: demand picks 0.01897 inside the band, in 37 solves. Green and blue end apart because they assume airlines pay different things. **Middle**: the active set, numbered by first appearance — only whether it changes matters. **Right**: the demand adjustment; its reaching zero certifies the answer is the unregularised one.

There are two loops, and β lives in the wrong one to matter. The market solves, given (D, p^0) , for q and a . The traffic loop then maps the price the market returns into a new D . At a fixed point of the *outer* loop the price the market returns *is* the price the demand was formed at, $p = p^0$; put that into (16) and $a/\beta = 0$, so $a = 0$. **The converged solve is a solve of the rigid programme at the converged demand.** Where supply is single-valued the reported price is the supply curve’s own value, exactly as it should be.

Off the fixed point it does a great deal, and that is its job. On a flat supply segment at cost c^* the solve picks $a = \beta(p^0 - c^*)$: the market anticipates that the traffic model is about to move demand, because the price it is about to quote differs from the one its demand was formed at, and it moves the quantity by its own estimate of that response before quoting. That is a **Newton step on the outer fixed-point equation**. β is a *preconditioner*, not an elasticity, and $e = e_0(\beta - d)/(s + \beta)$ is the statement of exactly that.

Measured, on the five-pathway hard case of §7, whose clearing sits on a flat segment in both regions — precisely the case the objection is about:

η	sweeps	λ^M region A	delivered	spread vs $\eta = 0.5$
0.25	<i>did not converge</i>	—	residual 0.105 after 900	—
0.50	91	0.08725278	0.07326695	—
1.00	59	0.08725278	0.07326695	5.7×10^{-11}
2.00	128	0.08725278	0.07326695	3.1×10^{-9}

A factor of four in β moves the answer by 3×10^{-9} — the MDA tolerance, not a modelling difference — while moving the work by a factor of two and, at $\eta = 0.25$, deciding whether there is an answer at all.

So the honest description is narrower than the title. “Quantity from supply, price from demand” is what happens **on a vertical segment**, where it is the mechanism — and §7 measures a case in which it charges 0.238 EUR/MJ for a fuel whose dearest producer charges 0.0997. Everywhere else the same machinery runs as an accelerator and the price still comes from supply.

It is not free. Off the fixed point the market has cleared $D + a$ while AeroMAPS’s energy budget is D ; `_reconcile` takes the mix from the market and the level from the budget, so an intermediate iterate carries a price computed at one quantity and volumes scaled to another. They agree at convergence — $|a|/D$ is 5.6×10^{-8} at $\eta = 0.5$ above — which means **the MDA tolerance is doing real work**: stop the loop early and the published price and the published volumes answer different questions. `demand_adjustment` is published so that this is checkable. No test asserts on it today.

Practical consequence: $w = 0$ was already solid. $w > 0$ is now a scenario dial rather than a research problem, at the cost of one parameter whose value affects the route and not the destination.

6.6 The last three cells: a tolerance below the solver’s own precision

After §6.5 three cells still failed — the whole stiffest-saturation column, $n = 16$, at $w = 1$. The natural guess is that the demand slope is mis-set there, and the linearised iteration supports it: writing the error propagation of the anchored step as

$$e = e_0 \frac{\beta - d}{s + \beta}, \quad d = \left| \frac{dD}{dp} \right|, \quad s = 1/S'(q), \quad (17)$$

a *stiff* supply curve drives $s \rightarrow 0$ and removes all tolerance for under-stating β . (Equation (17) also explains Table 6: at $\beta = d$ it gives $e = 0$, one-step convergence, which is why $\eta = 0.5$ beat $\eta = 1.0$ — the anchored scheme is a modified Newton method whose slope is set by hand.)

The guess was wrong, and the active-set signature said so at once.

The residual’s largest contributor is `region_B:hefa_fog_energy_consumption` at 2.87×10^4 MJ on volumes of $\sim 10^{13}$ — a relative 2.86×10^{-9} , which is exactly the kernel’s own `solver_tolerance` of 10^{-9} once scaled. The MDA was asking for 10^{-10} , one decade below the precision the market can produce. And it cannot produce more: at `solver_tolerance` 10^{-11} or 10^{-12} **Clarabel refuses the stiff cone outright** and the kernel raises rather than returning a worse answer.

A convergence tolerance is only meaningful against the precision of the disciplines underneath it. Coupling a conic solver into a fixed-point loop imports that solver’s noise floor into the loop’s stopping rule. The symptom — a run reporting non-convergence having in fact converged — is indistinguishable from a real failure unless the discipline itself is instrumented.

This is the third instance of the report’s recurring shape. §9: a programme that solved and was wrong. §6.4: a price that had no single value. Here: a loop that had arrived and was told it had not.

A finding about the code rather than the model. That tolerance was hard-coded at 10^{-10} in two places in `multi_regional_process.py` and could not be set from a scenario; the TODO beside it said as much. Assigning it on the chain after construction silently does nothing, which is how the first attempt at this measurement produced a false negative. It is now `regionalisation.mda_tolerance` (with `mda_max_iter`), defaulting to what it was, so nothing that ran before runs differently.

6.7 Price regimes across the obligation (measurement 4.5.2)

The brief’s third question is *how do prices behave when the obligation becomes unreachable?* The measurement it specifies is a multiplier on the obligation swept $0.5 \rightarrow 2$ at the inventory’s growth limit, recording the compliance price, the volume bought out, the delivered price and the traffic — so every point is a full two-region MDA.

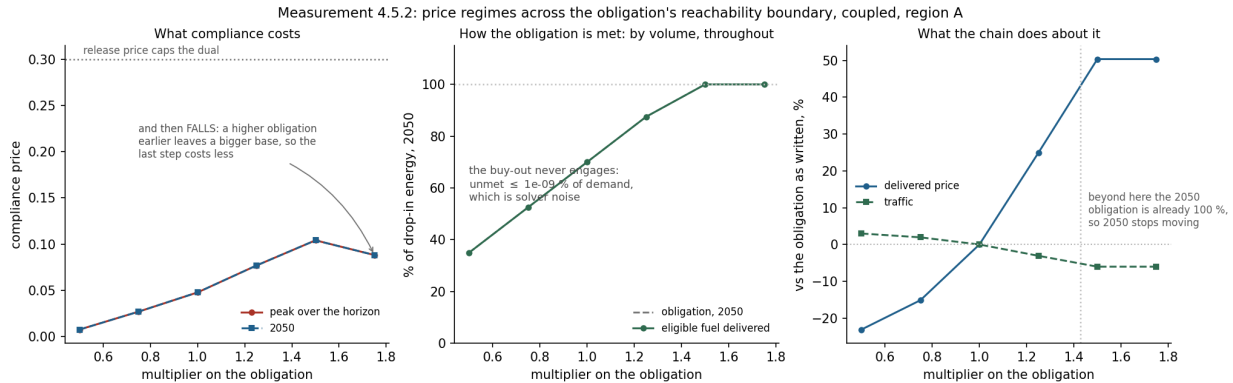


Figure 6: The obligation is met by volume throughout; the middle panel’s earlier version plotted the unmet volume on a 10^{-9} axis and drew a shape out of rounding error.

The buy-out never engages. At the inventory’s growth limit the obligation is reachable across the whole range the brief asks for — up to 100 % eligible fuel in 2050 — so the unreachable regime is never entered and **the brief’s third question is not answered by this sweep**. It is answered by the growth-limit sweep (§5), where at 15%/yr the buy-out engages exactly once, at the 2035 step. *Which instrument creates the shortfall matters:* tightening the obligation does not, because perfect foresight builds ahead of it; tightening the growth limit does.

An earlier version of this box reported that the compliance price “rises, then falls”, on a $\times 1.5$ versus $\times 1.75$ comparison. That comparison no longer exists: correcting the kerosene plumbing (§9.1) moved every number in this table and left $\times 1.5$ non-convergent. The remaining converged points are monotone. The claim is withdrawn rather than re-fitted to what survived — it rested on two adjacent points that differed only through a clipping artefact, which was thin evidence for a mechanism the report already demonstrates twice elsewhere (§8).

7 Five pathways: a curve that bends against one that jumps

This section answers a question the rest of the report could not even pose. Step 1 limits a pathway in one of two ways. Either its cost **bends** — γ and n make marginal cost rise as output approaches K — or its output is **capped**, and cost stays flat right up to the wall. Decision 5 chose the first because a hard cap gives a step-function dual and a step-function price whipsawed the coupling loop. §6.5 removed that failure at the source, so the cap is available again.

The two-pathway bench cannot distinguish them: with one sustainable pathway the obligation names the only fuel that can meet it, so there is no allocation to make and a staircase with one step has no step to step between. `scenario/energy_carriers_five.yaml` adds the real merit order, lifted verbatim from the default carrier data:

Both configurations use the same K , the same loose ramp-up, the same penalty, the same demand anchor and the same MDA tolerance. Only what K means differs. The hard ceiling did not exist before this measurement; it is new in the kernel as `capacity_limit`, with dual `capacity_price`, and carries seven tests including the stationarity identity of §5.2 (price = cost + capacity rent +

growth-limit rent, net of next year's), checked with the growth limit both slack and binding, and an independent objective score.

7.1 What it does, 2050, $w = 1$

The obligation is 70 % of drop-in fuel in *both* regions; region B's is larger in volume because its traffic is. Each region has its *own* 7.6 EJ/yr of capped fuels. Region A needs just past them — 7.8 EJ with hard caps, 8.3 with bends, which let the capped fuels stretch instead of calling on e-fuel. Region B needs 11.4 EJ, 3.8 of them from e-fuel whatever the supply shape.

Read the two halves of table 10 against each other. In region B the three configurations give the *same* λ^M to five decimals, the same delivered price, the same airfare, the same traffic and the same eligible volume — 11.385 EJ in all three. Only the mix differs, and with it the CO₂. In region A they differ by a quarter.

The difference between the regions is **which pathway is on the margin**. Region B's obligation is large enough to reach `electrofuel`, which is uncapped and unsaturated, so it sets the price at its own cost in every configuration — and how the pathways *below* it are limited cannot touch a price it alone determines. Region A's obligation stops inside the capped set, so the marginal unit is a limited one and the shape of its limit *is* the price.

The shape of a supply limit matters only where a limited pathway is marginal. Where an unlimited backstop is marginal, γ and n change who earns the rent and nothing else — not the price, not the airfare, not the traffic.

And where it does matter, it matters in one direction: **soft saturation understates scarcity**. That understatement is the whole gap between a 65 % rise in the compliance price and an 87 % one, and between 18.5 % and 24.1 % of traffic. $n = 16$ recovers most of the hard answer, which is the documented $n \rightarrow \infty$ limit arriving.

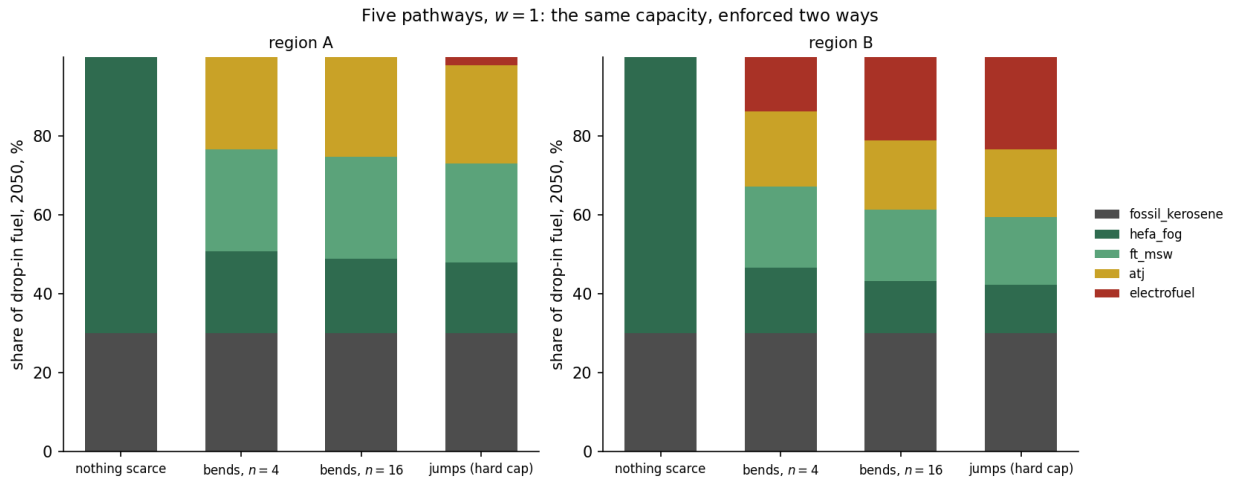


Figure 7: The 2050 mix. Region A never needs `electrofuel` under soft saturation; under a hard cap it does. Region B's price is set by `electrofuel` throughout, so only the split below it moves.

7.2 So: keep γ , n , K ?

Keep K . Drop γ and n as the default.

The argument is not that the bent curve is wrong. It is that at $n = 4$ the market charges 0.065 for the marginal eligible unit — a number that is not any pathway's cost, that no plant would quote, and that is an interpolation between 0.0397 (`atj`) and 0.0997 (`electrofuel`). **It is**

a price for a fuel nobody makes. The hard cap says instead that the next unit after `atj` runs out is an `electrofuel` unit at `electrofuel`’s cost — a statement an engineer can check, and disagree with.

γ and n also cannot be calibrated: there is no measurement whose answer is “ $\gamma = 1, n = 4$ ”. K can be. It is a plant fleet, in MJ/yr, and ENSPRESO and the ReFuelEU impact assessment have numbers for it.

And the reason for the soft form has expired. It was chosen because a hard cap whipsawed the loop. The hard-cap runs here converge in 91 Gauss–Seidel sweeps, against 44 for $n = 4$ and 97 for $n = 16$: the staircase is *not* the slowest of the three.

Keep γ and n available. They are the right tool for a pathway that genuinely has a rising cost curve rather than a wall — a feedstock drawn from a graded resource, where the next tonne really is dearer than the last. That is a *different* physical claim from “the plant makes 2 Mt/yr”, and the model should be able to say both.

7.3 The price/quantity plane, and where the price stops coming from cost

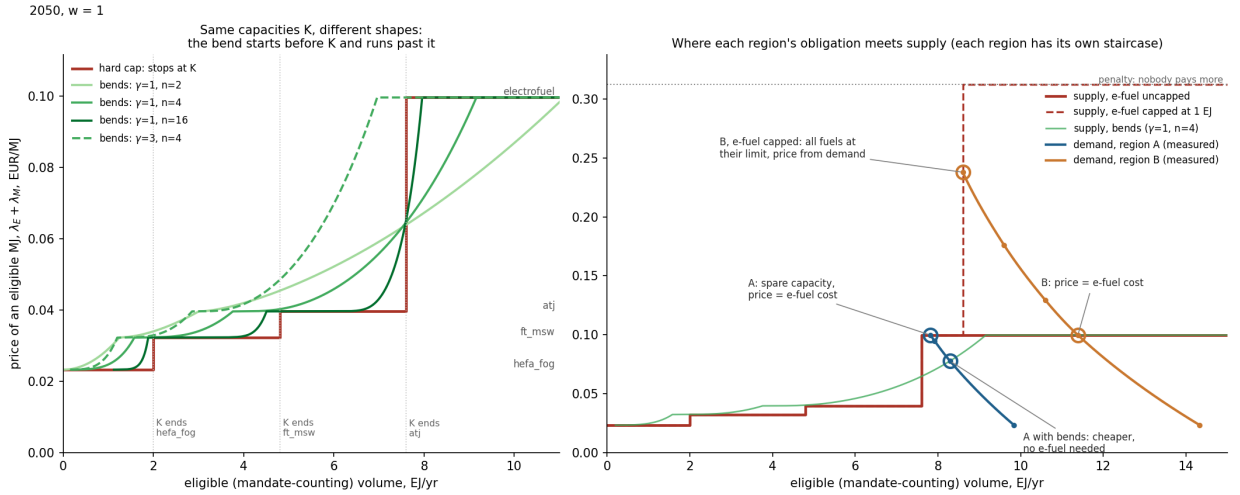


Figure 8: 2050, $w = 1$; x is eligible volume, y what an eligible MJ is paid, $\lambda^E + \lambda^M$. **Left**, supply alone: the same capacities K as a staircase and bent for several (γ, n) . **Right**: each region’s demand for eligible fuel crossing its own, identical, staircase. The demand curves are *measured* — each dot is one converged run — not assumed.

Left: why the bends come early. A bent pathway’s cost is already rising as it approaches K , so the next fuel becomes the cheaper one sooner — at $n = 4$, `ft_msw` enters at 1.58 EJ instead of 2.0, where $(q/K)^4 = c_{ft}/c_{hefa} - 1$ — and nothing stops it past K . Higher n tends to the staircase; higher γ bends harder and reaches the backstop sooner.

Right: the demand side is measured, not drawn. Every converged $w = 1$ run shares AeroMAPS’s traffic chain and differs only in supply, so each is one point (mD , $\lambda^E + \lambda^M$) of the *same* demand curve for its region; the lines join the points and nothing is fitted. Four extra runs move the crossing along the curves (e-fuel capped at 2 and 3 EJ; every capacity scaled $\times 0.8$ and $\times 0.65$). The last two did not converge (§14), so region B’s curve has five points and region A’s four. An earlier version of this figure drew demand as the market’s own linearisation, a line of slope $-m^2\beta$ — a tangent through the right point with an arbitrary slope, and so the wrong picture of demand.

The clearing point of the hard run sits at the *foot of the electrofuel step*, on a flat segment — not on a riser. That bounds the whole concern about step-function duals: **with an uncapped backstop in the set, the staircase never leaves the price undefined.** Some pathway is always marginal at its own cost. The vertical segment can only bite when every eligible route is

limited at once.

So the `all_capped` case builds exactly that: `electrofuel` capped too, at 1.0 EJ/yr, putting total eligible capacity at 8.6 EJ/yr. The traffic loop then walks demand down until the obligation exhausts it — and in one run, the two regions land in the two different regimes:

What the multipliers do when the price really does come from demand.

Region A sits on a flat segment: the eligible unit is paid exactly `electrofuel`'s cost to five decimals, and no multiplier exceeds a cost difference. λ^E is the residual's cost, λ^M is the marginal eligible cost minus it, and every inframarginal pathway's multiplier is the pure cost gap $\lambda_p^K = (\lambda^E + \lambda^M) - c_p$.

Region B sits on the vertical. Supply fixes the *quantity* — mD equals the capacity to the digit — and says nothing about the price. Equation (16) picks it, with $a \rightarrow 0$, so $\lambda^E + m\lambda^M = p^0$; the split is still pinned from the supply side because kerosene is still available, so

$$\lambda^M = \frac{p^0 - c_{\text{kerosene}}}{m}$$

The compliance price becomes a demand-side object: willingness to pay, net of the fossil cost, divided by the obligation share. Taking p^0 from the *previous* iterate — an independent number, not a rearrangement — this reproduces λ^M to 2.8×10^{-13} in region A and 8.8×10^{-13} in region B.

The consequence is the 0.23786 above: **2.4 times the most expensive thing anyone makes**, with 0.138 EUR/MJ of it above every cost in the model. That is not an error. It is the scarcity rent, and it is what a price does when the last unit cannot be produced at any cost. Traffic falls 46.9%.

Two things to be careful of. The $1/m$ factor *leverages* the compliance price: at $m = 0.7$ a 0.01 error in the delivered price is a 0.014 error in λ^M . And on a vertical segment β stops being a preconditioner and becomes the demand curve whose crossing sets the price — so that 0.238 should be read as “what this demand curve implies”, not as a forecast, until the coupled response is calibrated against the traffic model it stands in for.

Two costs. `all_capped` needs $\eta = 2$, not the bench's 0.5: on a vertical segment the loop's true response is stiff, and at $\eta = 0.5$ it does not converge in 900 sweeps. §6.5's rule — over-state β rather than under-state it — is here the difference between an answer and none. And at $w = 0$ the case does not converge at all (residual 0.09 after 900 sweeps), which is unexpected enough to record: $w = 0$ prices at average cost, so the loop should be *less* reactive. That is unexplained, and so are the two scaled-down probes above — three failures, all where every eligible route is at its limit.

7.4 What this measurement does not establish

- **The capacity levels are illustrative.** 2.0/2.8/2.8 EJ/yr were sized to put the obligation inside the capped set, not taken from a resource assessment. Every number in table 10 moves with them. The *structural* result does not: it turns on which pathway is marginal, not on where the caps are.
- **capacity_limit is one number per pathway, for all regions and years.** A real plant fleet grows; this one does not. Per-region and per-year ceilings are a grammar change, not a kernel change — the kernel already takes an (R, P, T) array.
- **Region A and region B differ in traffic growth too** (3.0 against 4.5 %/yr), so the contrast mixes two causes. The claim survives it — an identical λ^M across three configurations cannot be explained by growth — but the *sizes* are not like for like.

8 Policy cases

Five pathways — kerosene, HEFA from waste oil, HEFA from crop oil, alcohol-to-jet, e-fuel. *Kernel-level* (Table 2). Conclusions are about orderings and mechanisms; the numbers are illustrative.

8.1 Eligibility differing by region

One ReFuelEU-like region excludes crop feedstock; the other allows it; same headline obligation.

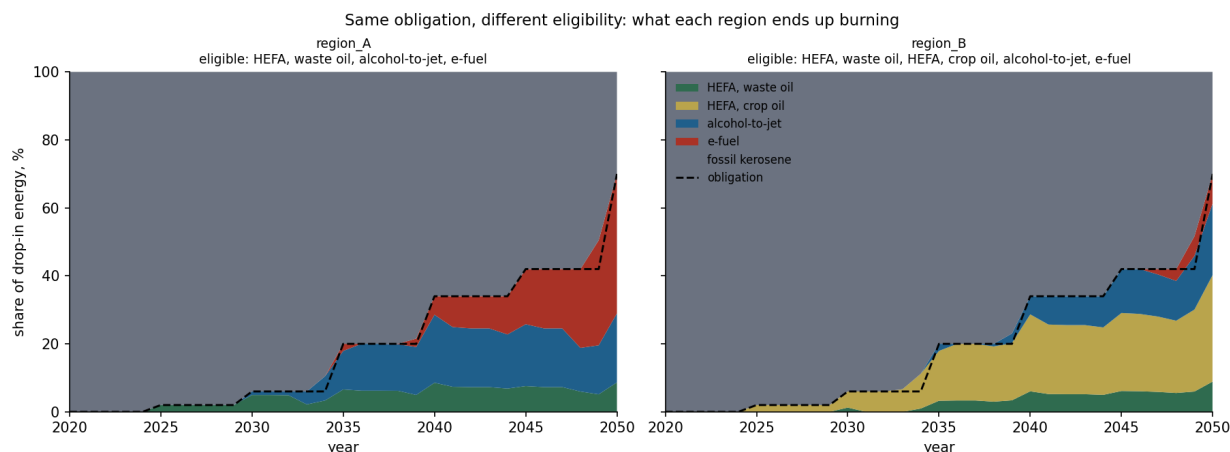


Figure 9: Same target, different portfolios. The coloured band is mandated fuel; the dashed staircase is the obligation, so colour above the line is anticipation.

Excluding a cheap feedstock is dearer every year except the last: +60 % on the compliance price in 2030, +139 % in 2040, and −10 % in 2050.

The mechanism is the growth limit. Denied the cheap option, region A builds e-fuel early and arrives at 2049 with a 30.9 % base; the permissive region coasts on crop oil and arrives with 5.0 %. When the obligation jumps 42 %→70 %, all four of the permissive region’s pathways hit their growth limits at once.

Cheap options early leave you worse placed for a steep step later — a result a share-allocation model cannot produce, because it has no notion of what was built when.

8.2 A sub-target within the target

ReFuelEU’s synthetic-fuel trajectory, on top of the headline obligation:

By 2050 e-fuel *exceeds* the 35 % target, because the forced early build-out leaves a bigger base to grow from.

A number that invites the wrong conclusion. The main compliance price falls in every year — by 53 % in 2040. That is *not* the sub-target making compliance cheaper: an extra constraint on the same programme can only make the optimum dearer, and it does. **Total production cost** +10.7%; **fuel bill** +5.1%. What falls is the price of *one instrument* because the other is now doing that work. Quoting the compliance price alone would show a policy paying for itself while it does the opposite.

8.3 A carbon tax, and nothing else

Every obligation switched off.

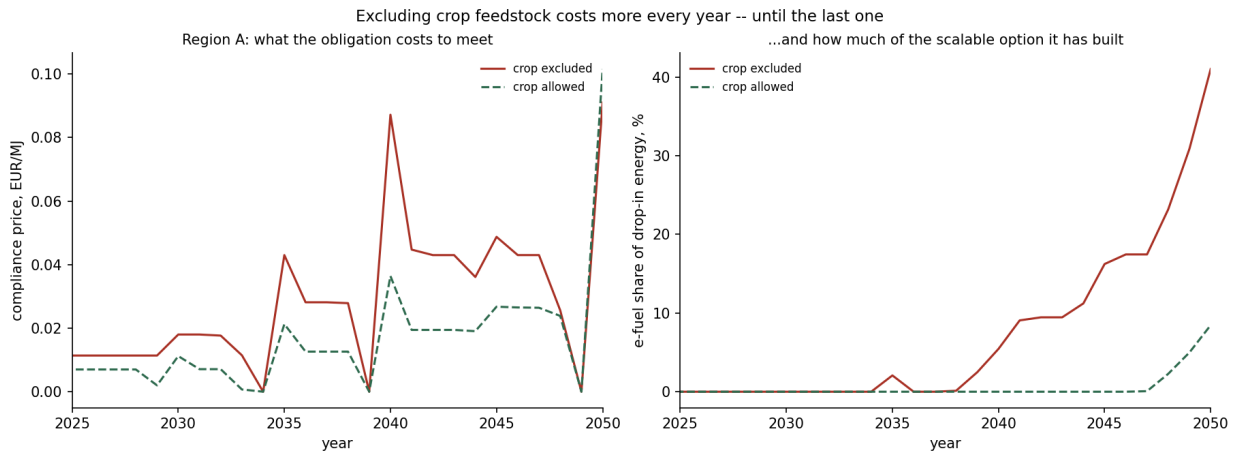


Figure 10: The cost, and the reason for it. Left: the exclusion is dearer in every year until the last; the sawtooth is each obligation step landing and the market catching up. Right: what it bought — a built-out e-fuel industry.

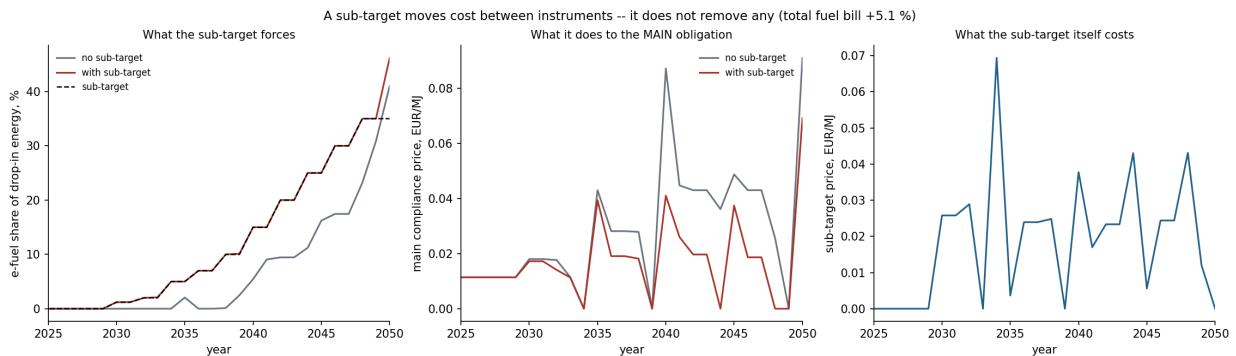


Figure 11: Left: the sub-target tracked exactly, then overshoot. Middle: the headline obligation gets cheaper throughout. Right: the sub-target's own price.

Below about 170 EUR/t a carbon tax on aviation fuel buys nothing at all. The mix stays 100 % kerosene and the intensity does not move from 88.6 gCO₂/MJ — it is pure revenue.

Then a cliff between 400 and 600, where kerosene goes from 67 % to zero and intensity falls 71.2 → 15.7. Not an artefact: kerosene's tax-inclusive cost ($0.0120 + 88.6\tau/10^6$) crosses e-fuel's ($0.0550 + 5.0\tau/10^6$) at $\tau = 514$ EUR/t, and the whole residual switches at once, because nothing distinguishes one litre of kerosene from another. **A tax moves one crossing point at a time, so its effect is a staircase, not a slope.**

8.4 The same destination by two instruments

The tax is **10.8 % cheaper** and **13 % dirtier**: it buys the cheapest sustainable fuel, the obligation's feedstock rule buys a cleaner one.

The environmental work is done by the eligibility rule, not by the volume target. A mandate written in volumes is not an emissions instrument. It becomes one only through the rules about what counts.

One confound: the tax case has no obligation, so no eligibility rule applies to it and crop oil is available. That is realistic — eligibility rules only exist inside mandates — but it makes this a *restricted mandate against an unrestricted tax*, not instrument against instrument at equal

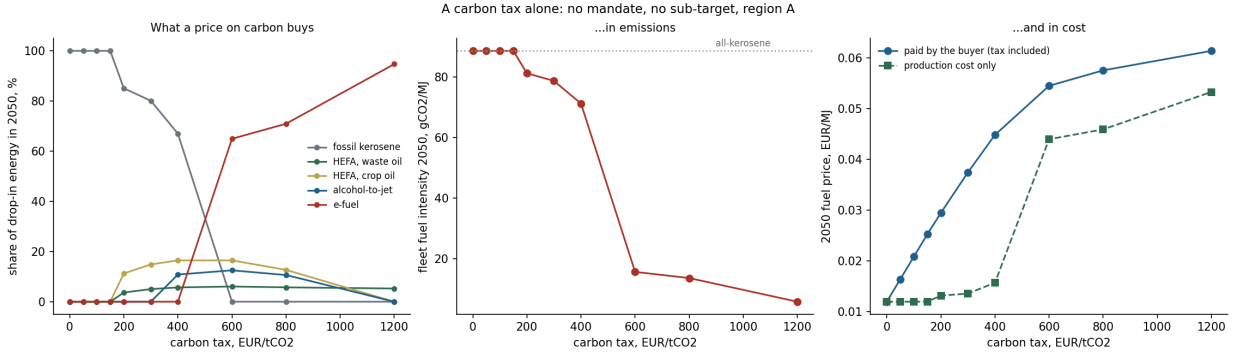


Figure 12: Nothing, then everything.

restriction.

9 Defects found, and one general lesson

Three defects of the *same family* were found, all of them producing plausible-looking numbers rather than errors.

1. **A price for an obligation that does not exist.** Where $m = 0$, constraint (3) is implied by $q, x \geq 0$ — redundant, so its multiplier is free. The solver reported up to 0.0112 EUR/MJ, roughly the whole cost of kerosene, in years with no obligation.
2. **An indeterminate split at a 100 % obligation.** There the obligation forces conventional supply to zero, so (2) and (3) have the same active rows and only their *sum* is determined. The solver split the identical problem three different ways, including a *negative* energy price.
3. **A jump at a binding/slack boundary** — §6.4. The same thing dynamically, and the one that stopped the coupled solve.

In every case a constraint stopped being independent and its multiplier became arbitrary. The first two are pinned to their meaningful KKT values. The third could not be pinned, because the discontinuity is real and neither branch is more meaningful than the other — it needed information the cost side does not contain, which is what §6.5 supplies.

9.1 A global setting that reached the wrong pathway

Found by an external review of this report, and present in *every* coupled measurement in it.

FuelClearing._pathway_setting resolves a per-pathway override if one is given and otherwise falls back to the global value — for every pathway, including the residual. All three coupled bench scripts pass `capacity` and `saturation_intensity` at the top level, meaning them for the pathway under study. So **kerosene was saturating too**, at about 90 % utilisation, and its marginal cost rose with output.

The consequences were not confined to kerosene. Re-running the whole grid with the residual exempted:

- $w = 0$ barely moves (at most -1.6 points, since the delivered price there is an average cost);
- $w = 1$ moves *up* by as much as **14 points** ($+99.6 \rightarrow +113.4$ at $\gamma = 0.5, n = 2$). Saturating kerosene raised λ^E but lowered λ^M by more, so it was suppressing the scarcity signal the measurement exists to show;
- measurement 4.5.2 lost two of its seven points and its “rises then falls” finding with them.

The residual is now exempt from the global scarcity settings and takes the no-scarcity default unless named explicitly under `pathways`: — a fossil supply limit is a legitimate thing to model,

it simply has to be asked for. Six tests in `test_fuel_clearing_discipline.py`, which is also the first test of the discipline rather than the kernel.

And the check on the fix turned up a second thing. With kerosene free, λ^E should equal the kerosene cost. It does not — it sits 3.6 % above it, by a gap that is *identical in every year and both regions*: 0.00044350.

That is exactly the carbon tax. $88.7 \text{ gCO}_2/\text{MJ} \times 5 \text{ EUR/tCO}_2 = 0.00044350$. So the identity does hold, on the *net* basis the programme decides on (§3.4) — and `fuel_market_energy_price` is published on that net basis while `{p}_market_mfsp` is published gross, with the wedge stripped. Two published prices, two bases, no note saying so. The duals are correct; the reporting is inconsistent, and anyone differencing them would find a carbon tax they did not put there. Recorded, not yet fixed: which basis a dual should be published on is a design decision, not a bug fix.

Two claims this report previously made and now withdraws

- **“Acceleration fixes the $w = 1$ failure.”** It was true of the one cell it was measured on. Run over the grid, Secant acceleration returned 3 cells of 12. It was a coincidence of trajectory, not a remedy: no sequence transformation can smooth a map that is genuinely discontinuous. The acceleration has been removed from the coupled grid.
- **“ $w > 0$ is a research problem, not a scenario dial.”** It was, until the failure was measured rather than inferred. It is now a dial (§6.5).

Both errors share a shape with the defects above, and with the four successive instrumentation attempts that each produced a confident and false reading of the coupled loop: *a measurement of one case, generalised*. The active-set signature exists so that this particular question can be answered by looking rather than by arguing.

A convex programme that *solves* is not a programme that is *right*. The saturation term was first written as $a q^{n+1}$ with $a = c\gamma/((n+1)K^n)$ — algebraically correct, numerically unusable: a carries K^{-n} and the coefficients spanned 35 orders of magnitude at $n = 16$. Half the parameter grid refused to solve; worse, *where it did solve it returned a suboptimal point*, 0.13 % dearer, which showed up as a **3.4 % error in the delivered price** — at the exact settings the whole sensitivity table was built on. The solver reported **optimal** both times.

Rewriting it in the utilisation it is actually a function of, $[c\gamma K/(n+1)] (q/K)^{n+1}$, made all 36 cells solve. The only thing that caught the wrong answer was scoring the returned point against the objective *independently of the solver*.

10 Limitations

- **In the market, regions still do not interact.** Every constraint of `FuelClearing` is per region and the objective a plain sum, so solving jointly and separately agree to solver noise: no **leakage**, no competition for a feedstock. A global pool now exists in the kernel (§12) and flows are carried through AeroMAPS (§13), but the two are not yet joined.
- **One planner, with perfect foresight,** stands in for every agent: no banking of compliance, one discount rate for a fuel bill and a plant, every mandate fully credible (§5.2). Kept deliberately (§11).
- **$w > 0$ needs a demand slope to converge** (§6.5). It does not affect the answer where supply is single-valued, but too small a value costs 200 sweeps or the run. It should be derived rather than configured — or made unnecessary by putting demand inside the kernel (§11).

- **The MDA tolerance has to be chosen against the solver’s precision** (§6.6). The market mode runs at 10^{-7} ; the default stays 10^{-10} . Nothing warns a user who pairs them badly.
- **The elastic balance is a local linearisation**, exact at the fixed point and only there: enough to select a price, not enough to be a demand model.
- **The buy-out payment reaches no one** (§2.5).
- **No learning curve** anywhere in AeroMAPS, so every price dynamic comes from the model’s own scarcity terms.
- **Bottom-up costs are excluded “for now”**: a vintage *mix* makes average cost *fall* with volume and the cost integral concave. Choosing vintages rather than reading a mean removes the obstacle (§11).
- **2050 is the horizon** and also the steepest step, so results turning on “what is built by the end” are partly a terminal condition.
- **Emission factors, capacities and four of the five costs are assumed.**

11 Step 2: decisions taken, options still open

Step 2 is taken one diagnosed decision at a time, not as a full-scale model. The purpose every option is judged against: **estimate what airlines will pay for fuel under a variety of policies worldwide, and the effect on traffic growth; eventually tune those policies; and see how regions interact through a globalised fuel market.**

11.1 Decisions

11.2 Market and traffic: put demand inside the kernel?

The problem. The kernel is a supply-side market; traffic lives in AeroMAPS, and the two meet through the Gauss–Seidel loop. That loop is where most of this report’s difficulty came from: price intervals at kinks (§6.5), a demand slope that must be over- rather than under-stated, tolerances tied to the solver’s precision (§6.6), and the failures of §7.

What the traffic block actually is. In `models_operation_cost_top_down_feedback` it is already a small market in equilibrium — a linear airline supply curve (`PassengerAircraftMarginalCost`) crossing an iso-elastic passenger demand (`RPKElasticity`):

$$\begin{aligned}
 A &= a_t \text{RPK} + b + (C_t^{\text{nf}} - C_0) + e_t p + \tau_t && \text{airline supply} \\
 \text{RPK} &= \text{RPK}_t^0 (A/A_0)^\varepsilon && \text{passenger demand, } \varepsilon = -0.9 \\
 E &= e_t \text{RPK} + E_t^{\text{other}} && \text{fuel demand}
 \end{aligned}$$

with $a_t = 2(A_0 - C_0)/\text{RPK}_t^0$, $b = 2C_0 - A_0$, e_t in MJ per RPK and RPK_t^0 the traffic before any price effect. Both integrate into terms a convex solver accepts. Passenger value $U_t(\text{RPK}) = A_0(\text{RPK}_t^0)^{-1/\varepsilon} \text{RPK}^{1+1/\varepsilon}/(1+1/\varepsilon)$ is concave (exponent -0.111 here, a power cone); airline cost is quadratic. One programme can then hold fuel supply, airline supply and passenger demand at once, maximising value minus all costs. Its optimality condition in RPK is AeroMAPS’s airfare equation,

$$U'_t(\text{RPK}) = a_t \text{RPK} + b + C_t^{\text{nf}} - C_0 + \tau_t + e_t(\lambda^E + m \lambda^M),$$

with the fuel term equal to the single market price of \$2.5. The elastic balance of §6.5 is the first-order version of this: η is the slope of a tangent to a curve the kernel could hold exactly.

Moves when the fuel price moves → solved inside	Does not move → an input, computed upstream
fuel price, fuel mix, airline costs and airfare, traffic, energy demand, fuel bill	RPK^0 (traffic before any price effect), non-fuel costs, MJ per RPK, taxes, base-year calibration, each pathway’s cost

The left column is exactly the loop the MDA iterates on today. With demand inside, one solve finds all of it together, so **that loop no longer needs an MDA**: AeroMAPS runs its chain once afterwards, and a zero residual confirms the two agree. The right column does not depend on the fuel price, so it arrives as ordinary inputs, with no dry run.

Why average-cost pricing does not fit this option. The solve does not compute a price and hand it to demand: it chooses fuel volumes and traffic *together*. At the optimum the last passenger-km is worth what it costs to serve, and that cost is a derivative — the marginal price $\lambda^E + m\lambda^M$. That is what an optimum is, not a rule added to it. Average cost can still be imposed, as a constraint tying traffic to $\sum_p c_p q_p / \sum_p q_p$; but that is a ratio of the solve’s own variables inside a nonlinear equation, and the problem stops being convex — with it go the single certified answer and the multipliers as prices. Average cost is better served by today’s loop, which at $w = 0$ converges easily (16 solves, figure 5) because the average cost moves smoothly.

It is also the less natural assumption: when supply is short, every seller gets the market price, and paying average cost describes contracts or regulation. **So: $w = 1$ with demand inside, one solve; $w < 1$ in today’s loop mode**, with average cost and producer rents reported as outputs either way.

What else it costs. Only some demand models fit: RPKElasticity yes; logistic_income probably (a multiplicative price index, unverified); the fleet-push model no, since deliveries make energy demand depend on the traffic *path* — those keep today’s loop. The airline calibration is duplicated in the kernel, guarded by the one-sweep test. It forces a pending AeroMAPS choice: whether the carbon tax sits inside or outside the airline supply function. And whether freight energy responds to price is to be checked. **Decided by D1–D2.**

11.3 Supply: a capacity variable, using bottom-up data

Soft saturation invents prices and cannot be calibrated (§7); a fixed cap is a plant fleet that never grows; and growth limits on production put anticipation on burning fuel early (§5.2). The option: decide **capacity additions by vintage**, production $\leq$ installed capacity $\times$ load factor, growth limits on additions.

How it gets its costs. Each vintage — the plants commissioned in a given year — has its own unit cost: that year’s capex annualised, plus opex and resources. The kernel computes these numbers itself before the solve, with the formula and inputs `BottomUpCost` uses (capex, opex, lifespan, construction time, load factor, private discount rate), then decides how much of each vintage to build. `BottomUpCost` is not run inside; fed the chosen build-out afterwards through `{p}_energy_production_commissioned`, which it already reads, it reports the mean costs and should agree. Decision 6’s objection does not arise, because the programme chooses vintages rather than reading a mean. Pathways documented only top-down need one extra number, the capital share of their cost.

It adds steps (one per vintage), not smoothness; what it adds is *time* — rents appear while plants are short and pay for the next ones. Capex falling with the commissioning *year* fits; capex falling with *cumulative* capacity (endogenous learning) is non-convex and does not. **Decided by D3.**

11.4 Policy instruments: from one mandate to a policy mix

Most of the gap is plumbing and linear constraints. The one structural step is the discipline carrying a **list of obligations per region** and publishing one price per obligation (§2.5). Without shared supply, instruments in different regions still cannot affect each other — which is the pool’s job.

11.5 Regions through one global pool

Each region **supplies** the pool with what it produces — its plants and feedstock, where production-side policies such as per-litre credits act — and **draws** what it consumes, where use-side policies such as mandates, eligibility and fuel taxes act. The market arbitrates; each region’s net flow is tracked, without routes or transport costs. That is the mechanism the objective is about: a mandate in A pulls in the cheap feedstock, and B pays more for its fuel. Its volume side exists in the kernel (§12); its plumbing through AeroMAPS has been checked with an explicit stand-in for the market (§13). **Its price side is D5.**

11.6 Tuning policies

Tuning is an **outer optimisation** over policy parameters with the market as the inner model — a GEMSEO scenario, not an MDA; an MPEC in the terms of §11.7. It needs no change to the market’s foresight. Sensitivities exist on each smooth piece of the solution and only as subgradients where the active set changes, so designs of experiments or derivative-free methods are the safer first choice. With demand inside the kernel the outcomes vary continuously with policy, which makes this better behaved. **Measured by D4.**

11.7 MCP, MPEC, EPEC

Optimisation — what the kernel is. One objective, one solve; its optimality conditions *are* the equilibrium, whenever that equilibrium is “as if” one planner minimised total cost.

MCP — mixed complementarity problem. Every agent’s optimality conditions and the market-clearing conditions written directly, as pairs “either this quantity is zero, or its margin is”: $x \geq 0$, $F(x) \geq 0$, $x F(x) = 0$. Every convex optimisation can be written so — that is its KKT system — but an MCP also covers equilibria no objective gives: price-dependent support, recycled revenue, agents with different discount rates or beliefs. Cost: existence and uniqueness are not guaranteed, and the solvers (PATH) need a reasonable start.

MPEC — one leader optimising against an equilibrium: the regulator chooses a mandate, the market responds. This is policy tuning.

EPEC — several leaders, each an MPEC, in equilibrium with one another: every region setting its own policy while anticipating the global market and the others’. Non-convex; may have no equilibrium or several. Research grade.

Neither MCP nor EPEC is needed now. Keeping the kernel’s optimality conditions explicit — as the identity tests do — keeps a later move to an MCP a reformulation, not a rewrite.

11.8 Diagnostics, in order

Before D1, one scoping choice: which traffic models must work with demand inside — constant elasticity only; plus logistic-income; plus fleet-push?

12 A first step: one global pool in the kernel

The volume side of the pool only: each region supplies it with what it produces and draws what it consumes, and the flows are tracked. How pooling moves each region’s *price* is D5’s question and is deliberately not answered — the prices are recorded, not interpreted.

12.1 What the kernel now builds

One new, optional input, a flag per pathway (pooled). Without it the kernel builds exactly the programme it built before. With it, a region’s two sides separate:

Side	Variables	Constraints that act there
produces	$q_{r,p,t}$, as before	cost, saturation, hard cap, growth limit — plants and feedstock
consumes	$d_{r,p,t}$ for pooled pathways	energy balance, obligations, eligibility — airlines and policy
pool	—	$\sum_r q_{r,p,t} = \sum_r d_{r,p,t}$ per pooled pathway and year

A pathway not flagged stays local ($q = d$). New outputs: production, the net flow (production minus consumption; it sums to zero over regions exactly) and the pool price (the multiplier on the pool balance). Two identities hold and are tested: at every region *producing* a pooled fuel, pool price = cost + capacity rent + growth-limit rent, net of next year’s; wherever a region *draws* it, its delivered marginal price for that fuel is the pool price. Costs are read as the *producing* region’s; a cost levied where fuel is burnt — a carbon tax on use — has no place yet.

12.2 Flows are not unique, so a second solve picks the least trade

Two regions making the same fuel at the same cost with capacity to spare can split the pool’s supply any way at the same total cost, and an interior-point solver returns the middle of that set: each region producing for the other for no reason — 35.1 EJ traded over 2020–2050 where the pool needed 1.7. So a pooled solve is followed by a second one that fixes the optimal cost as a constraint and minimises the exported volume. The prices stay those of the first solve, which is exact: in a convex programme every optimal primal point pairs with every optimal dual point. It costs a few milliseconds. Pool prices of a fuel nobody draws are an interval, and are pinned to its lower end — the same remedy as the compliance price without an obligation.

12.3 The bench

Kernel only, at **fixed demand** taken from the converged hard-cap run of §7, so traffic does not respond to pooling. The separate case, re-solved, returns that run’s volumes to 3.3×10^{-8} of demand; an independent year-by-year LP agrees with the kernel’s cost to 1.3×10^{-7} in every year.

Case	Plants	Traded 2020–50	E-fuel made	Discounted cost
separate	HEFA 2.0, FT 2.8, ATJ 2.8 EJ/yr in each region	—	4.0 EJ	7.1047×10^{12}
pooled	same	1.7 EJ	4.0 EJ	−0.08 %
separate, waste oil in A	HEFA 4.0 in A, 0 in B	—	16.4 EJ	7.4777×10^{12}
pooled, waste oil in A	same	36.0 EJ	4.0 EJ	−5.1 %

- **Same plants in both regions:** trade happens in 2040–2044 only, while A still runs on FT-MSW and B would otherwise climb to ATJ. In every other year both sit on the same marginal fuel and nothing needs to move.
- **Waste oil all in A:** separately, B never gets a litre of it and climbs to e-fuel from 2042. Pooled, B draws A’s surplus from 2025, the resource is **used up from 2035** instead of 2040, and four times less e-fuel is made.
- **Where the plants sit stops mattering to the totals:** the two pooled cases have the same cost to 4×10^{-10} . Without transport costs, relocating a plant changes who exports, nothing else — the decision of §11 showing, not a defect.

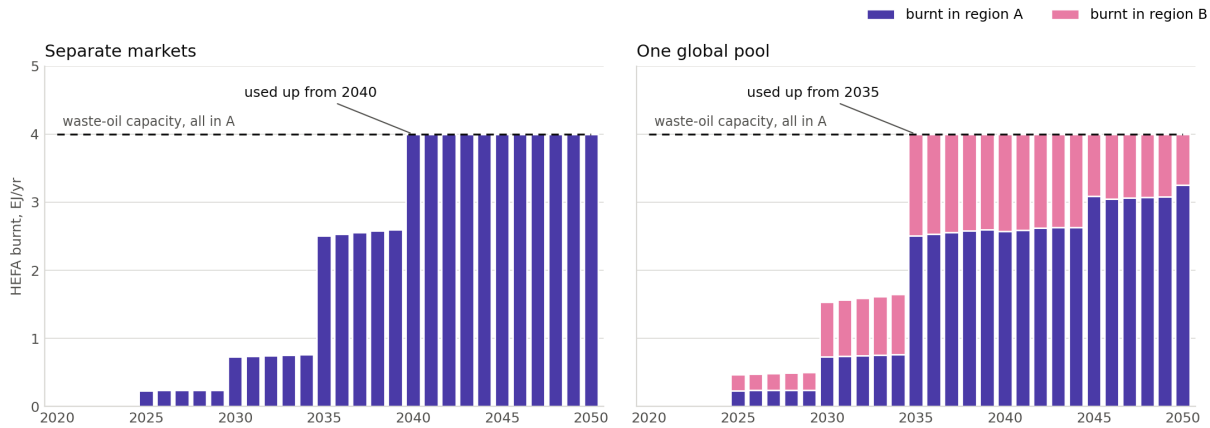


Figure 13: HEFA burnt per year when all the waste-oil capacity is in region A. Separately, A alone uses the resource up only in 2040; pooled, B draws on it from 2025 and it is used up from 2035.

12.4 What the pool determines, and what it does not

The least-trade pass pins **how much** each region exports, not **which fuel**. Once an exporter runs two fuels at their caps, both earn a rent and both sell at the price of the marginal fuel above them, so shipping either costs the same. Measured directly — the smallest and largest net flow of each fuel over all points optimal in cost *and* in traded volume: with only HEFA capped (waste oil in A, 2025–2039) every flow is determined to the MJ; from 2040, A exports 2.42 EJ/yr and its HEFA share can be anything from 0 to 2.42.

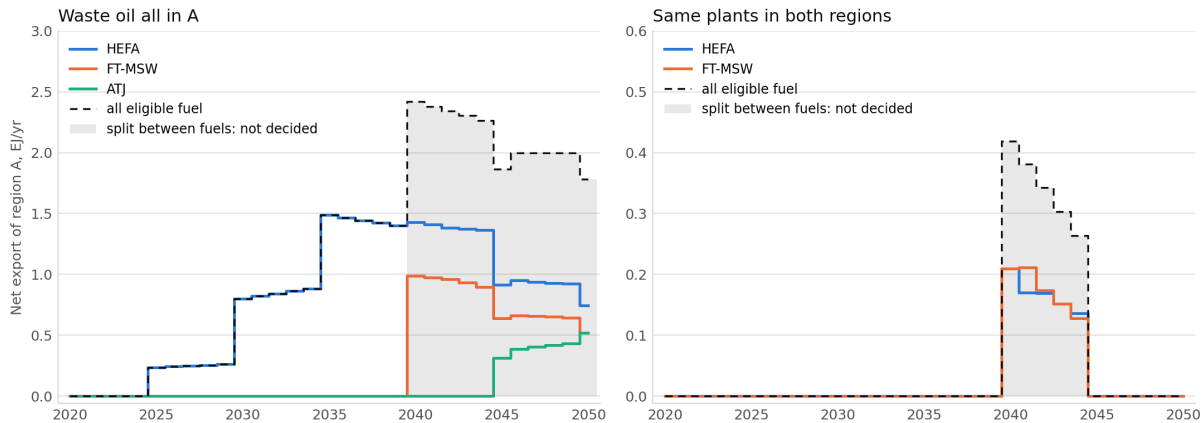


Figure 14: Region A's net exports by fuel. The dashed total is determined; the grey area marks the years where its split between fuels is not.

This matters: which fuel a region burns decides the emission factor its CO_2 carries and, at $w < 1$, its average price; and an arbitrary split in a coupled loop is a quantity that can flicker between sweeps, which is the failure of §6.5. Differing use-side carbon taxes or eligibility rules would break these ties by themselves (read off the stationarity conditions, not yet measured); a uniform carbon tax would not.

Decided. The market will say which fuel is traded, once it has the logic to — use-side costs among it. Until then an **explicit matrix** stands in for the market's flow logic, and the first thing to establish is whether the multi-regional plumbing carries flows at all. The carbon tax goes to the consumer side; how far that moves the market is for later.

13 Plumbing for flows between regions: an explicit matrix

No market logic here. The flows are stated by hand, as a sourcing matrix, and the only question is whether the multi-regional process carries them: a region producing what it does not burn, each side reaching the models that belong to it, every MJ accounted for once. The kernel’s pool is not used; it is the market logic meant to replace the matrix later.

13.1 What was added

Piece	What it does
FuelTrade (global model)	Reads every region’s <code>{p}_energy_consumption</code> , applies the matrix, emits each region’s <code>{p}_energy_production</code> and net export, and every tracked flow. Prospective years only.
<code>fuel_trade</code> flag (under regionalisation)	Tells every region that production is no longer consumption; same route as <code>fuel_market</code> .
Resources follow production	Feedstock use computed on production ; CO ₂ stays on consumption . Off by default.
Guards	Flag and model declared together or not at all; <code>unified_mda</code> only; bottom-up pathways refused under trade.

The matrix, per pathway, gives each consuming region the shares it sources from each producing region; each row sums to one, and anything not listed is made where it is burnt. On the bench, all the HEFA region B burns is made in A, and a quarter of A’s kerosene is made in B.

Nothing structural changed: each region is still a full, namespaced AeroMAPS process in one MDChain, and FuelTrade joins FuelClearing among the global models. The two new couplings — consumption to production, production to the aggregator — are feed-forward, so trade adds no loop: the MDA takes the same number of sweeps with and without it.

13.2 The checks

Four full two-region runs on the five-pathway bench — standard mode and market mode, each without and with trade. Worst relative error, the same in both modes:

Check	Result
Everything a consumer sees unchanged: consumption per pathway, CO ₂ , airfare, RPK, delivered fuel price (17 series × 2 regions)	0, bit-identical
Production is what the matrix says	0
World production = world consumption, per pathway and year; also through the aggregator	2×10^{-16}
Tracked flows add up to each region’s net export	3×10^{-16}
Feedstock = $1.14 \times$ production, in each region	10^{-11}
World feedstock use unchanged	10^{-16}
Historical years: production = consumption	0

What moves, 2050: region A burns 9.8 EJ of HEFA and makes 24.2; B burns 14.3 and makes none. A’s waste-oil use goes from 11.2 to 27.6 EJ and B’s to zero: B’s demand now exhausts A’s resource. Fourteen tests, including one end-to-end MDA run, each checked against a deliberate breakage (feedstock booked on consumption again; the matrix transposed).

13.3 What this does not do yet

- **Nothing reads production except feedstock.** Costs are still each region’s own: B pays B’s cost for HEFA made in A — identical on this bench, not in general. This is where trade first touches prices, deferred with the rest of the price interaction.

- **With the market on, caps would act on consumption**, since `FuelClearing` clears separate regional markets. A real combination needs the pool of §12 to decide the flows — which is its point.
- Bottom-up pathways are refused under trade, and the matrix is constant in time.

14 Open items

1. **Join the pool to the plumbing.** The pool is in the kernel, the flows are carried through AeroMAPS, the discipline still clears separate markets. Before wiring: a use-side cost input, and a rule for which fuel is traded where the market leaves it open.
2. **The step-2 diagnostics** D1–D5 (§11.8), D1 first.
3. **One obligation per region in the discipline;** the sub-mandate and regional eligibility are kernel-only, and λ^S is computed but not published.
4. **Three undiagnosed non-convergences**, all where every eligible route is at its limit: `all_capped` at $w = 0$, and capacities scaled $\times 0.8$ and $\times 0.65$ at $w = 1$.
5. **No test asserts that the demand adjustment vanishes at convergence** — the guard on the price/volume consistency the discipline’s reconciliation depends on.
6. **capacity_limit is one number per pathway**, for every region and year.
7. **Two settings a user can set badly with no warning:** the demand slope and the MDA tolerance against the solver’s.
8. **The buy-out payment** does not enter the cost chain.
9. **cvxpy as a proper optional dependency** before the mode ships.

Symbol	Shape, unit	Meaning and provenance
$D_{r,t}$	(R, T) , MJ	Drop-in energy demand. In coupled mode this is <code>energy_consumption_dropin_fuel</code> , an output of the fleet chain — which is what makes the market part of a loop.
$c_{r,p,t}$	(R, P, T) , EUR/MJ	Production cost per unit energy at zero utilisation. Taken as <code>{p}_net_mfsp</code> : the cost the buyer actually faces, carbon tax and subsidies included (§3.4). Flat in time on the bench — AeroMAPS has no learning curve, so all price dynamics come from the model, not the supply curve.
$m_{r,t}$	(R, T) , fraction	The obligation: minimum eligible share of demand. A <i>fraction</i> , not a percentage — validated, because a silent factor of 100 here would be invisible downstream.
$\mathcal{S}_r$	(R, P) , boolean	Eligibility , not chemistry: which pathways count towards region r ’s obligation. The same fuel may count in one jurisdiction and not another.
$\mu_{r,t}, \mathcal{N}_r$	$(R, T), (R, P)$	Sub-obligation and its narrower eligible set — ReFuelEU’s synthetic-fuel target. Optional; absent by default.
$B_{r,t}$	(R, T) , EUR/MJ	Penalty (“buy-out”) per unit of unmet obligation. Caps λ^M : nobody pays more to comply than to not comply. Without it an unreachable obligation makes the programme <i>infeasible</i> rather than expensive.
$K_{r,p,t}$	(R, P, T) , MJ	Capacity <i>scale</i> , not a cap: the output at which the pathway starts to strain. ∞ disables saturation. Exogenous at step 1.
$\gamma_{r,p}$	(R, P)	Saturation intensity — how much dearer the next unit is at full utilisation. 0 switches the term off entirely.
n	scalar ≥ 0	Saturation stiffness — how <i>abruptly</i> the extra cost arrives. $n \rightarrow \infty$ recovers a hard cap, which is what the soft form exists to avoid.
$g_{r,p}$	(R, P) , /year	Growth limit: relative expansion allowed year on year.
$s_{r,p}$	(R, P) , MJ	Growth <i>seed</i> . Not optional in practice: with $q^{\text{init}} = 0$ and $s = 0$, (5) pins the pathway at zero forever. Not a free dial either — it is Eq. (12)’s volume branch (§2.6).
$q_{r,p}^{\text{init}}$	(R, P) , MJ	Production in the last historical year; the growth limit’s initial condition.
δ_t	scalar	Discount factor $(1 + \rho)^{-t}$. Decides how far ahead of a step it is worth building.
w	scalar $\in [0, 1]$	Pricing weight . Decides who keeps the scarcity rent (§2.5). The single most consequential parameter.

Table 1: Inputs. R regions, P pathways, T prospective years.

Symbol	From	Reads as
$\lambda_{r,t}^E$	(2)	What the next unit of energy costs. Set by whichever pathway is on the margin.
$\lambda_{r,t}^M$	(3)	What one more unit of obligation costs. Bounded above by B . Zero in any year where the market is already ahead of the obligation — a slack constraint costs nothing.
$\lambda_{r,t}^S$	(4)	The same, for the narrower target.
$\lambda_{r,p,t}^R$	(5)	What it would be worth to be allowed to grow faster. Zero where the limit is not in the way.

	Kernel only	Chained into AeroMAPS
What runs	The convex programme alone. NumPy arrays in, arrays out. No AeroMAPS import.	The programme as a GEMSEO discipline inside the full multi-regional MDA.
Demand $D_{r,t}$	Fixed from outside — taken from a stored reference run.	An output of the model , responding to the price the market sets.
Loop closed?	No. Price has no effect on demand.	Yes: price $\rightarrow$ operating cost $\rightarrow$ airfare $\rightarrow$ elasticity $\rightarrow$ demand $\rightarrow$ price.
Cost per case	~ 0.03 s	minutes
Used for	Formulation, verification, parameter sweeps, all policy cases .	Reproduction end-to-end, convergence, the coupled grid.

Table 2: The two levels. Everything in §5 and §8 is kernel-only; everything in §6 is coupled.

Parameter	Swept	Spread in delivered price
Pricing weight w	$0 \rightarrow 1$	37.3 %
Stiffness n	$1 \rightarrow 16$	16.3 %
Penalty B	$0.02 \rightarrow 0.30$	8.3 %
Growth limit g	$0.10 \rightarrow 1.20$ /yr	3.8 %
Intensity γ	$0 \rightarrow 4$	1.7 %
Discount rate ρ	$0 \rightarrow 0.08$	0.0 %

2050, both regions	reference	market	worst rel., all years
hefa_fog_energy_consumption	9.84171e12	9.84171e12	1.1e-09
hefa_fog_share_dropin_fuel	70	70	1.9e-09
dropin_fuel_mean_mfsp	0.019819	0.019819	1.8e-10
dropin_fuel_mean_co2_emission_factor	41.1	41.1	1.0e-09
co2_emissions_passenger	493.04	493.04	6.3e-10
airfare_per_rpk	0.0867286	0.0867286	5.1e-11
rpk	1.98233e13	1.98233e13	1.6e-11

Table 3: Worst relative difference across every variable and year: 1.9×10^{-9} .

	no scarcity	scarce, $w = 0$	scarce, $w = 1$
Delivered price 2050	0.019819	0.021839 (+10.2 %)	0.04020 (+103 %)
Max compliance price	0.01083	0.047886	—
RPK 2050	1.98233e13	−1.2 %	−10.8 %
CO ₂ 2050	493.04	−1.2 %	—

	$n = 2$	$n = 4$	$n = 8$	$n = 16$
$w = 0, \gamma = 0.5$	+10.7/ − 1.3	+5.2/ − 0.6	+1.8/ − 0.2	+0.4/ − 0.0
$w = 0, \gamma = 1$	+21.0/ − 2.4	+10.1/ − 1.2	+3.6/ − 0.4	+0.8/ − 0.1
$w = 0, \gamma = 2$	+40.5/ − 4.6	+19.5/ − 2.2	+7.1/ − 0.8	+1.6/ − 0.2
$w = 1, \gamma = 0.5$	+113.4/ − 11.8	+91.0/ − 9.7	+73.5/ − 8.0	+64.4/ − 7.1
$w = 1, \gamma = 1$	+138.8/ − 14.1	+115.4/ − 12.0	+83.6/ − 9.0	+66.9/ − 7.3
$w = 1, \gamma = 2$	+183.8/ − 17.8	+138.8/ − 14.1	+100.5/ − 10.6	+71.4/ − 7.8

Table 4: **24 of 24** cells converge, on plain Gauss–Seidel with *no* acceleration, against 15 of 24 before §6.5. Both rows are monotone in n and in γ ; the irregular pattern §6.4 once tried to explain was never a pattern, only a trend read off whichever cells happened to survive. The last three needed a second and unrelated diagnosis (§6.6).

	market solves	converged?	distinct active sets	changes, 2nd half
$w = 0$	23	yes	5	0
$w = 1$, rigid balance	202	no	6	33
$w = 1$, demand-anchored	60	yes	6	0

Table 5: The failure, watched rather than deduced. At $w = 0$ the set of tight constraints stops moving and stays stopped. At $w = 1$ it never stops.

Device	solves	converged?	$ a /D$ final	
none (rigid)	202	no	—	the failure
volume anchor alone, $\rho = 0.5$	202	no	—	cannot help: the volume is pinned
$\eta = 0.05$	202	no	2.8×10^{-3}	slope too small, overshoots
$\eta = 0.2$	202	no	1.5×10^{-9}	market settled, chain had not
$\eta = 0.5$	59	yes	6.5×10^{-11}	
$\eta = 1.0$	150	yes	2.5×10^{-10}	over-damped
$\eta = 3.0$	202	no	2.9×10^{-5}	over-damped past the iteration budget

Table 6: Ablation at $w = 1$, two regions, no acceleration. The fastest η is an estimate of the chain’s own elasticity of drop-in demand to the delivered fuel price: the iteration is fastest when the slope it is told matches the slope it faces.

$n = 16, \gamma = 0.5, w = 1$	active-set changes	tail price step	$ a /D$ final
$\eta = 0.5$	0	8.3×10^{-11}	5.8×10^{-10}
$\eta = 2$	0	7.5×10^{-11}	2.6×10^{-7}
$\eta = 8$	0	1.3×10^{-6}	1.2×10^{-3}

Table 7: The market has converged. Raising η only makes it worse.

$n = 16, \gamma = 0.5$	MDA tolerance 10^{-10}	MDA tolerance 10^{-8}
outcome	failed, residual 2.863×10^{-9}	converged
delivered price 2050	—	0.032553681
$n = 8$ control	0.034053429	0.034053429

Table 8: The control is the point: a cell that converges either way returns a *bit-identical* answer at the looser tolerance. This loosens the stopping rule, not the result.

multiplier	2050 obligation	λ^M	bought out	delivered 2050	RPK 2050	eligible share
$\times 0.50$	35 %	0.04005	0	0.015980	2.027×10^{13}	35.0 %
$\times 0.75$	52.5 %	0.05660	0	0.018374	1.999×10^{13}	52.5 %
$\times 1.00$	70 %	0.05951	0	0.021824	1.959×10^{13}	70.0 %
$\times 1.25$	87.5 %	0.07784	0	0.027277	1.899×10^{13}	87.5 %
$\times 1.50$	100 %	<i>did not converge</i> (1.84×10^{-7})				
$\times 1.75$	100 %	0.08831	0	0.032832	1.841×10^{13}	100 %
$\times 2.00$	100 %	<i>did not converge</i> (1.46×10^{-6})				

Table 9: Coupled, region A. The obligation clips at 100 % above $\times 1.43$, so every 2050 quantity stops moving beyond that point.

Pathway	Net cost 2050, EUR/MJ	Capacity used here
fossil_kerosene	0.01244	residual, uncapped
hefa_fog	0.02327	2.0 EJ/yr
ft_msw	0.03234	2.8 EJ/yr
atj	0.03966	2.8 EJ/yr
electrofuel	0.09970	uncapped — PtL is bound by electricity and capital, not by a feedstock

Region A	λ^E	λ^M	delivered	airfare	RPK	CO ₂	eligible
nothing scarce	0.01244	0.01083	0.01982	—	—	—	9.84 EJ
bends, $n = 4$	0.01244	0.06526	0.05787	+25.5 %	−18.5 %	−0.3 %	8.29 EJ
bends, $n = 16$	0.01244	0.08222	0.06974	+33.5 %	−22.9 %	−4.6 %	7.92 EJ
jumps	0.01244	0.08725	0.07327	+ 35.9 %	− 24.1 %	−6.4 %	7.82 EJ
Region B							
nothing scarce	0.01244	0.01083	0.01982	—	—	—	14.33 EJ
bends, $n = 4$	0.01244	0.08725	0.07328	+35.9 %	−24.1 %	−10.7 %	11.385 EJ
bends, $n = 16$	0.01244	0.08725	0.07328	+35.9 %	−24.1 %	−12.1 %	11.385 EJ
jumps	0.01244	0.08725	0.07328	+35.9 %	−24.1 %	−12.5 %	11.385 EJ

Table 10: The same capacities, enforced two ways, through the full two-region MDA. Percentages are against the same run with nothing scarce.

2050, $w = 1$	eligible/cap	λ^E	λ^M	paid	dearest	rent above all
region A	7.818/8.6 spare	0.01244	0.08725	0.09970	0.09970	−0.00000
region B	8.600/8.6 full	0.01244	0.22542	0.23786	0.09970	+ 0.13817

	2030	2040	2050
e-fuel, no sub-target	0.0 %	5.4 %	41.0 %
e-fuel, with sub-target	1.2 %	15.0 %	46.2 %
Main compliance price	0.01800 → 0.01721	0.08718 → 0.04103	0.09100 → 0.06908

Same 70 % sustainable, 2050	Production cost	Intensity	Mix
Obligation (crop excluded)	0.03427	36.5	41 % e-fuel, no crop
Carbon tax, 515 EUR/t	0.03056	41.3	33 % e-fuel, 18 % crop

Decision	Why
Keep one convex optimisation , not separate optimisations for airlines, producers and planners	With price-taking agents, convex costs and shared information, separate optimisations linked by market clearing reach <i>exactly</i> the solution of the single programme (the welfare theorems) — slower, with more loops. They would not remove price intervals either: those belong to the market. Separate agents only pay off for market power, strategic regulators or differing beliefs (§11.7).
Perfect foresight , no myopic mode	Policy planning is framed with foresight, as in the bilevel optimisation work. The anticipation it produces is penalty avoidance; the instrument it acts through is what needs fixing (§5.2).
Regions interact through one global pool ; no routes, no transport costs	Each region supplies the pool with what it produces and draws what it consumes; the market arbitrates and each region's net flow is tracked. Long-run target: a shared <i>feedstock</i> pool, so that policy A and policy B interact through it.
Instruments beyond Re-FuelEU are in scope	Carbon taxes, subsidies and their interaction across regions must be first-class.
Plumbing before market logic , and diagnose before implementing	Each new mechanism is first carried through AeroMAPS with an explicit stand-in and checked end to end; the market logic comes after. Each open option has a measurement that decides it (§11.8).

Instrument	Example	Today	Fits one optimisation?
Share mandate + penalty	ReFuelEU, UK SAF mandate	yes, one per region	yes
Sub-mandate	ReFuelEU synthetic sub-target	kernel only; λ^S not published	yes — plumbing
Eligibility differing by region	feedstock caps, exclusions	kernel only	yes — plumbing
Carbon tax, as a given price	national carbon taxes	yes, via pathway costs	yes
Emissions trading	EU ETS, intra-EU flights	yes, as a given price	yes: aviation is a small part of the cap, so its price is effectively given. Only a cap on aviation <i>alone</i> makes the price an output.
Fixed subsidy or tax per MJ	per-litre production credits	yes, via pathway costs	yes
GHG-intensity standard	California LCFS, German GHG quota	no	yes: one linear constraint
Subsidy with a fixed budget	national SAF funds	no	yes: its multiplier prices the money
Support whose amount depends on the market price	a guaranteed strike price topping up the gap (UK revenue certainty mechanism); penalty revenue recycled	no	no : costs may depend on quantities, but prices are the optimisation's outputs — an MCP (§11.7)

	Diagnostic	It decides	Effort
D1	Rebuild market + airlines + passengers as one problem in a scratch script, from converged runs; compare with the MDA	Whether demand inside is exact, or which AeroMAPS term it misses	~1 day
D2	D1 at $w = 0$ and 0.5	What partial rent pass-through costs once demand is inside	$\frac{1}{2}$ day
D3	Capacity-variable prototype on one pathway, top-down split vs bottom-up data	Whether anticipation moves onto plants; how rents evolve	1–2 days
D4	Fine sweep of the mandate level, today vs demand inside	Whether outcomes are smooth enough to tune policies	$\frac{1}{2}$ day
D5	Two regions on one global pool, different mandates and carbon taxes	How far one region's policy moves the other's price	1 day